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**RECOLOR: A MOBILE VISION ASSISTIVE TOOL FOR COLOR VISION
DEFICIENCY USING ADAPTIVE DALTONIZATION
AND CLUSTERING ALGORITHMS**

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Bachelor of Science in Computer Science

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CIE	Commission Internationale de l'Éclairage (International Commission on Illumination)
CNN	Convolutional Neural Network
CVD	Color Vision Deficiency
FCM	Firebase Cloud Messaging
FSIM	Feature Similarity Index
GPU	Graphics Processing Unit
HSV	Hue, Saturation, Value
ISO/IEC 25010	International Standard for Software Quality Evaluation
LMS	Long, Medium, Short (cone response–based color space)
MARS	Mobile Application Rating Scale
ms/frame	Milliseconds per frame (latency unit)
NPU	Neural Processing Unit
NVSA	National Vision Screening Act

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PERI	Philippine Eye Research Institute
RA	Republic Act
RGB	Red, Green, Blue
SSIM	Structural Similarity Index
TFLite	TensorFlow Lite
UI	User Interface
UX	User Experience
WCAG	Web Content Accessibility Guidelines

CHAPTER I

INTRODUCTION

Background and Rationale of the Study

Color vision deficiency (CVD), affecting approximately 350 million individuals worldwide (1 in 12 males and 1 in 200 females; Asare et al., 2024), represents a pervasive yet under-addressed challenge in human-computer interaction and daily living. Recent epidemiological studies highlight regional disparities, with Africa reporting a pooled CVD prevalence of 2.71% (2.13% male, 0.34% female; Asare et al., 2024), while the United States reports 3.7% prevalence (7% males, 0.4% females; National Eye Institute, 2023). In the Philippines, studies show that approximately 5.17% of male high school students, particularly in Metro Manila, are affected, predominantly by the deutan type, which impairs green color perception (Cruz et al., 2010). Individuals with CVD face practical difficulties in tasks such as interpreting color-coded information, distinguishing traffic signals, and selecting appropriate clothing, often affecting confidence, independence, and access to opportunities.

In response to these challenges, technological solutions have emerged to complement traditional diagnostic tools such as the Ishihara Test and Farnsworth D-15. Software-based interventions and mobile applications, in particular, have shown potential to enhance accessibility by applying adaptive filters and real-time image processing techniques. However, most available tools remain limited to static images, lack context-aware adaptability, and fail to provide sustained functional assistance in dynamic, everyday environments. These limitations emphasize the

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	need for innovative solutions that combine diagnostic and corrective functionalities within an accessible mobile platform tailored to diverse user needs. The Philippine Eye Research Institute (PERI), established under Republic Act 4593, serves as the country’s leading institution for blindness prevention and eye research. In line with Republic Act 11358, or the National Vision Screening Act, PERI collaborates with the Department of Health (DOH) and the Department of Education (DepEd) to implement vision screening programs for Filipino children. While these initiatives ensure early detection of visual impairments, they currently emphasize refractive errors and amblyopia and lack dedicated screening or functional support for CVD. This reveals a critical research and implementation gap in addressing color vision deficiencies within the Philippine context, particularly in ensuring both detection and practical assistance for affected individuals. The proposed system, ReColor, seeks to address this gap by developing a mobile vision assistive tool that integrates adaptive daltonization, K-means clustering, and convolutional neural networks (CNNs). Unlike conventional screening tools, ReColor not only embeds the Ishihara Test for preliminary detection but also provides real-time color correction tailored to anomalous trichromacy, specifically protanomaly and deuteranomaly, which are the most prevalent CVD types among Filipinos. By combining diagnostic screening, functional enhancement, and data collection into a single mobile framework, ReColor aligns with the objectives of the National Vision Screening Act while extending its impact to daily living. This integration positions the system as a culturally responsive, accessible,	
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and scalable solution for improving quality of life and advancing inclusive design in the Philippines.

Objectives of the Study

This study seeks to design, develop, and evaluate ReColor, a mobile vision assistive tool for individuals with color vision deficiency. The application integrates adaptive daltonization, clustering algorithms, and convolutional neural networks (CNNs) to enhance real-time color perception, while also embedding Ishihara-based screening for early detection and awareness. It aims to serve as an affordable, scalable, and context-appropriate solution that supports both functional assistance and informed career guidance for affected individuals in the Philippines.:

1. To design and implement an adaptive color correction pipeline that integrates:
 - a. K-Means clustering for efficient grouping of pixels and reduction of computational overhead.
 - b. Daltonization algorithms for remapping colors along confusion lines to improve perceptual discriminability.
 - c. Convolutional Neural Networks (CNNs) for real-time color detection tailored to user-specific deficiency profiles.
2. To incorporate multimodal assistive features (haptic and auditory feedback) that enhance user accessibility and provide non-visual support for color distinctions.
3. To integrate a career-awareness component that informs individuals with CVD about professions requiring normal color perception to support informed career choices.

4. To evaluate the technical performance of the system using:
 - a. Color Difference (ΔE , CIEDE2000) to measure perceptual accuracy of corrected images.
 - b. Structural Similarity Index (SSIM) to ensure naturalness and structural integrity of enhanced images.
 - c. Processing Latency (ms per frame) to assess real-time performance and responsiveness on mobile devices.
 - d. Clustering Purity/Inertia to evaluate the efficiency and quality of K-Means clustering in color segmentation.
 - e. Cross-device benchmarking across representative Android devices to assess device compatibility.
5. To assess the acceptance, accessibility, and software quality of ReColor among color vision-deficient users and expert evaluators such as eye doctors and software specialists, based on Android app core standards, WCAG 2.2 accessibility guidelines, and validated statistical methods such as paired t-test and Cronbach's Alpha.

Significance of Study

This study holds significant value for multiple stakeholders, with the **Philippine Eye Research Institute (PERI)** as its foremost beneficiary. As the national center for ophthalmic research and a long-standing advocate of blindness prevention in the Philippines, PERI can utilize ReColor as both a reference tool and a potential digital platform for color vision deficiency (CVD) screening and management. By extending PERI's research focus beyond mainstream eye diseases such as cataract and glaucoma toward CVD, the study contributes to more inclusive visual health initiatives. Moreover, the application's adaptive framework may provide PERI with new avenues for public health interventions, education campaigns, and cross-agency collaborations aimed at improving visual accessibility for Filipinos.

Individuals with Color Vision Deficiency. By providing an affordable, real-time tool that dynamically enhances color discrimination and offers haptic and audio feedback, ReColor empowers users to navigate color-dependent tasks more effectively in education, employment, and daily activities. In addition to functional assistance, the application integrates early awareness and self-screening features such as the Ishihara test and career-awareness prompts. This allows undiagnosed individuals and young users to better understand the presence and impact of their condition, while also providing guidance on career paths that may be affected by CVD, promoting informed decision-making and long-term preparedness.

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	Educators, Healthcare Professionals, and Caregivers. This study also benefits educators who require inclusive teaching materials and methods that accommodate CVD learners, healthcare professionals who depend on accurate color-coded diagnostics, and caregivers who support the daily functioning of individuals with CVD. By introducing awareness components alongside real-time assistive feedback, ReColor facilitates improved communication, task assignment, and health monitoring, contributing to better outcomes and easing associated burdens. Proponents and Researchers. For the proponents and researchers, this study cultivates competencies in advanced image processing, accessibility standards, real-time system optimization, and sustainable design principles. It also strengthens expertise in human-centered design and public health informatics by addressing both functional assistance and early-awareness education. The project thus provides a unique opportunity to contribute interdisciplinary knowledge that bridges technology, healthcare, and social impact, enhancing academic credentials and professional skills in mobile accessible computing. Future Researchers and Innovators. This research lays a foundational open-source platform, inviting further exploration into adaptive color vision aids, career-awareness tools, and immersive technologies. It encourages future studies to refine algorithms, expand support to other forms of color vision deficiencies, and integrate career counseling frameworks with emerging assistive systems. By documenting methodologies and results, this study catalyzes continued innovation and cooperative development aimed at equitable assistive solutions worldwide.	
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	Scope and Limitation This research specifically targets of anomalous trichromacy prevalent in the Filipino population: Protanomaly (reduced sensitivity to red light) and Deuteranomaly (reduced sensitivity to green light). Together, these red-green color vision deficiencies affect approximately 6% to 8% of Filipino males, consistent with global trends stemming from X-linked inheritance patterns (Cruz et al., 2010; Pilestone, n.d.; Apollo Hospitals, 2025). The study excludes Tritanomaly (blue-light sensitivity deficiency) due to its rarity both globally and locally. By focusing on Protanomaly and Deuteranomaly, the project maximizes relevance and user impact given the high prevalence of these CVD types in the Philippines. The project scope restricts device compatibility low to midrange smartphones, reflecting the commonly available hardware among Filipino users. This ensures wide accessibility while maintaining sufficient performance capabilities to support real-time image processing and artificial intelligence (AI) inference required by the application. The system’s implementation and evaluation are confined to the Philippine setting, carefully considering local demographic, technological, and socioeconomic factors to tailor the application effectively. Evaluation will include usability testing, performance measurement, and expert validation from ophthalmologists and optometrists to ensure both technical accuracy and practical applicability. User scope will be limited to individuals of legal age, with testing conducted in both indoor and outdoor environments to account for varied lighting conditions that affect vision correction performance. The timeframe for development, testing,	
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	and evaluation is set between August 2025 and February 2026. Key technologies used include React Native (with react-native-vision-camera, react-native-reanimated, react-native-fs) for future cross-platform mobile app implementation, OpenCV for efficient real-time image processing, and TensorFlow Lite for AI model deployment on-device. In direct support of Republic Act No. 11358, or the National Vision Screening Act, the system also incorporates an awareness and early detection component through embedded Ishihara testing and career-awareness prompts. This feature not only recognize in potentially identifying CVD cases but also supports the Act’s objectives of early vision screening, research, and public health awareness. By providing guidance on careers that may be affected by CVD, ReColor extends the scope of national screening efforts toward long-term inclusivity, informed decision-making, and equitable participation in education and employment. Limitations include device variability affecting performance, reliance on manual input without sensor integration in the current phase, and absence of cloud or web-based management solutions, emphasizing a localized and resource-appropriate approach to address Filipino users’ needs and infrastructure realities.	
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CHAPTER II
REVIEW OF RELATED LITERATURE

Color Vision Deficiency Overview

Color vision deficiency (CVD) is a common visual impairment characterized by the reduced ability or inability to distinguish certain colors, primarily caused by anomalies or absence of cone photoreceptors in the retina responsible for color perception (National Eye Institute, 2025; American Optometric Association, 2021). The most prevalent form is congenital red-green color blindness, inherited in an X-linked pattern and affecting predominantly males (EnChroma, 2022). Other types include blue-yellow deficiencies and complete color blindness, though these are rarer. Additionally, acquired CVD can result from eye diseases, neurological disorders, or exposure to toxins (American Optometric Association, 2021). Globally, an estimated 8% of males and 0.5% of females experience CVD, which significantly impacts daily activities and occupational tasks reliant on accurate color discrimination (National Eye Institute, 2025; Colormax, 2021).

The functional limitations of CVD pose substantial challenges in education, healthcare, transportation, and employment, necessitating effective adaptive technologies (EnChroma, 2021). Current solutions, such as static optical filters and software with limited real-time adaptation, often fail to accommodate changing lighting conditions or individual severity of deficiency (Almutairi & Alharbi, 2022; Machado et al., 2023). These shortcomings underscore the critical need for dynamic, accessible, and affordable interventions to enhance color accessibility and improve

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	quality of life for individuals with CVD worldwide (Shinohara & Wobbrock, 2023; World Health Organization, 2020). . CVD Prominence in the Philippines In the Philippines, studies on the prevalence of Color Vision Deficiency (CVD) remain limited in scale. Regional evidence suggests rates comparable to international figures, with recent meta-analyses reporting prevalence among males ranging from 5% to 8%, with the deutan type being the most common (Almustanyir, 2025; Jeong, Kim, & Park, 2025). This aligns with broader global estimates but underscores the lack of nationwide, gender-inclusive prevalence data in the Philippine context. Globally, approximately 2.8% of the population—or over three million Filipinos—are affected, emphasizing the importance of early detection and heightened awareness (World Health Organization, 2020). Traditional diagnostic tools such as the Ishihara plates and Farnsworth D-15 remain widely used, while the anomaloscope, considered the gold standard, sees limited use locally due to high costs and technical requirements (Garcia, Martinez, & Fernandez, 2022; Kumar, Patel, & Singh, 2023). Despite these figures, CVD continues to be underrecognized in the Philippines, with awareness often occurring only when students face challenges in academic tasks that rely on color-coding, such as maps, charts, and laboratory work (Chen, Wu, & Zhao, 2023). The absence of systematic screening not only impacts educational performance but also constrains career opportunities, as individuals with CVD may face exclusion from fields requiring precise color discrimination such as	
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	engineering, aviation, and healthcare (Shinohara & Wobbrock, 2023). While Republic Act 7277, or the Magna Carta for Disabled Persons, provides broad protections for individuals with disabilities, CVD is not consistently acknowledged under this law, resulting in minimal policy-level accommodations (Department of Social Welfare and Development [DSWD], 2019). These limitations in research, policy, and inclusivity highlight the urgent need for accessible, technology-driven interventions tailored to Filipino users with CVD. ReColor as Contributory to the National Vision Screening Program (RA 11358) Republic Act No. 11358, or the National Vision Screening Act, calls for the development of a National Vision Screening Program (NVSP) for kindergarten students. The NVSP is developing early screening, referral, and research with a view to reducing visual impairment affecting learning and development and is currently being initiated by the Department of Education (DepEd) and the Department of Health (DOH) in collaboration with the Philippine Eye Research Institute (PERI). Trained teachers are assessing kindergarten students using standardized screening kits for conditions such as refractive error and amblyopia (PERI, 2019). Early screening helps reduce the burden of visual disability to benefit children in the long-term. Early screening supports the constitutional provision to promote health, along with inclusive education found in DepEd policies. ReColor contributes to the objectives of RA 11358 by embedding Ishihara based screening for Color Vision Deficiency (CVD) and extending beyond detection through real-time adaptive correction using daltonization and clustering algorithms. COLLEGE OF INFORMATION TECHNOLOGY AND COMPUTER SCIENCE	
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This mobile, accessible platform supplements the NVSP by addressing visual challenges not fully covered with traditional kits, particularly those involving red–green deficiencies, while ensuring scalability in resource-constrained settings (Khan, Li, & Chen, 2022). Furthermore, ReColor’s capacity to generate prevalence and usage data aligns with the law’s intent for continuing research and evidenced-based policymaking, allowing PERI and allied agencies to expand vision health strategies to historically underserved populations. Through a combination of screening, functional assistance, and research, ReColor serves as a technology-driven complement to the NVSP in support of the law’s more extensive agenda of improving health for all Filipinos at minimum-effects levels.

Different Types of Color Vision Deficiency

Table 1

Different Types of Color Vision Deficiency (CVD)

CVD Type	Description	Subtypes	Prevalence	Implications
Monochromacy	Complete absence of functional cone cells, or only one cone type functional, causing total color blindness (Cole, 2016; Dans et al., 2017).	Monochromacy (achromatopsia); Cone monochromacy.	Very rare: 0.00003%–0.003% globally; ~0.02% of Filipino children (Almutleb & Doshi, 2020).	Severe light sensitivity; limited acuity; no effective corrective aids available.
Dichromacy	Absence of one cone type, leading to two-dimensional color perception (Cruz et al., 2010; Wikipedia, n.d.).	Protanopia (no red cones); Deuteranopia (no green cones); Tritanopia (no blue cones).	~5.17% among Filipino males (mainly deuteranopia); tritanopia <0.01% and not sex-linked (Cruz et al., 2010).	Strong confusion between reds/greens or blues/yellows; high impact on academic/professional tasks.

Anomalous Trichromacy	All three cone types are present but one has altered spectral sensitivity, usually red-green (Neitz & Neitz, 2017).	Protanomaly (altered red sensitivity); Deuteranomaly (altered green sensitivity); Tritanomaly (rare, blue-yellow).	Most common: affects ~5% of males, especially deuteranomaly (Pattie et al., 2022).	Colors appear washed out; may partially benefit from optical aids, but costly and limited (Almutairi & Alharbi, 2022).
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Color Vision Deficiency (CVD) encompasses a spectrum of disorders characterized by the reduced ability or complete inability to perceive colors accurately due to anomalies in the cone photoreceptor cells of the retina. The severity and nature of CVD depend on the type and functionality of the affected cones, resulting in distinct types—monochromacy, dichromacy, and anomalous trichromacy—that vary in prevalence and impact on daily life (Neitz & Neitz, 2017; Cruz et al., 2010).

Monochromacy represents the most severe and rare form of CVD, where individuals either lack all functional cone cells (rod monochromacy or achromatopsia) or have only one type of working cone cell (cone monochromacy), causing complete color blindness and reliance on rod cells for vision (Cole, 2016; Dans et al., 2017). This condition, affecting approximately 0.00003% to 0.003% globally, severely limits visual acuity and light sensitivity, and current corrective options like color filters are ineffective due to the absence of cone function (Almutleb

& Doshi, 2020). In the Philippines, achromatopsia is noted to affect about 0.02% of children, highlighting a small but significant group for whom conventional assistive technologies offer limited benefits.

Dichromacy is a moderate form of CVD characterized by the absence of one of the three cone types, resulting in a two-dimensional color perception system. The three dichromatic subtypes include protanopia (lack of red-sensitive cones), deuteranopia (lack of green-sensitive cones), and tritanopia (lack of blue-sensitive cones), each presenting specific color confusions and perceptual deficits (Wikipedia, n.d.; Cruz et al., 2010). Deuteranopia is the most prevalent, particularly among Filipino males, with estimated prevalence rates around 5.17%, followed by protanopia, while tritanopia remains rare (less than 0.01%) and is not sex-linked (Cruz et al., 2010; Wikipedia, n.d.). Given the higher prevalence of dichromacy and its substantial impact on color discrimination tasks, it remains a critical target for assistive technology development.

Anomalous Trichromacy is the most common form of CVD, where all three cone types are present, but one has altered spectral sensitivity, primarily affecting red-green perception (**protanomaly and deuteranomaly**) with rare occurrences in the blue-yellow spectrum (**tritanomaly**) (Neitz & Neitz, 2017; Cruz et al., 2010). This condition causes colors to appear washed out, leading to confusion in distinguishing reds, greens, browns, and certain blues and purples, with deuteranomaly being the most frequent subtype, affecting approximately 5% of males (Pattie et al., 2022). Unlike dichromacy, anomalous trichromacy may partially respond to optical aids such as spectral filtering glasses; however, these solutions

are costly, static, and limited in effectiveness, particularly under varying lighting conditions (Pattie et al., 2022; Almutairi & Alharbi, 2022).

Comparative Analysis of Existing tools and methods for Color Enhancement

Table 2

Comparative Analysis of Methods Addressing Color Vision Deficiency

Method	Mechanism	Advantages	Limitations	Status
Optical Aids (Glasses/Contacts)	Filters overlapping wavelengths to enhance contrast (e.g., EnChroma, tinted lenses) (Almutairi & Alharbi, 2022).	Commercially available; immediate use; enhances subjective discrimination (Machado et al., 2023).	Lighting-dependent; high cost; does not restore normal vision (Almutairi & Alharbi, 2022).	Widely available but limited effectiveness.
Software/Digital Aids	Digital filters applied to images or video to separate confusing colors (Shinohara & Wobbrock, 2023).	Affordable; accessible; adaptable to devices (Shinohara & Wobbrock, 2023).	Often non-real time; limited beyond static images; slower on mobile devices (Machado et al., 2023).	Available as mobile/desktop apps.
Gene Therapy	Introduces functional opsin genes into cones to restore photopigments (Mancuso et al., 2009).	Targets root cause; potential to restore red-green perception (Mancuso et al., 2009).	Experimental only; ethical/safety concerns; not used clinically.	Tested in animal models only.

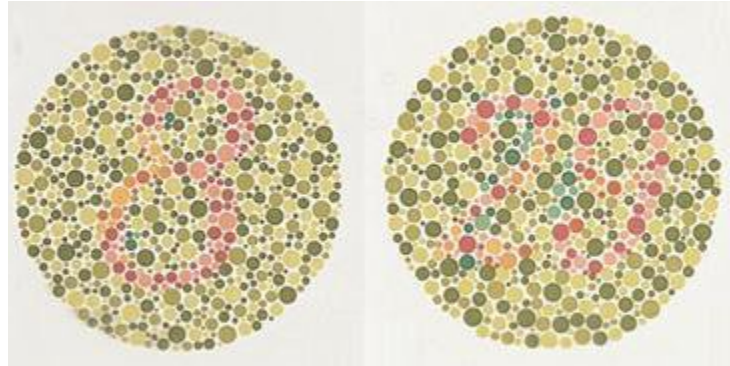
Color Vision Deficiency (CVD) remains a persistent challenge, with existing interventions offering varying degrees of effectiveness. Optical aids such as

EnChroma and VINO glasses filter overlapping wavelengths to enhance color contrast, showing some benefit for individuals with mild to moderate red-green deficiencies. However, their high cost, dependence on lighting conditions, and limited effectiveness for severe cases constrain their accessibility, particularly in low-resource contexts (Almutairi & Alharbi, 2022; Pattie et al., 2022). Software-based solutions, including digital filters and mobile applications, provide more affordable and widely accessible alternatives by modifying color outputs on electronic displays. Despite these advantages, many remain confined to static images, exhibit slow or non-real-time performance on mobile devices, and lack integration into dynamic daily tasks (Machado et al., 2023; Shinohara & Wobbrock, 2023). Emerging approaches in real-time digital processing aim to overcome these limitations through adaptive algorithms, though performance variability across devices continues to pose challenges. Meanwhile, gene therapy research has demonstrated groundbreaking results in animal models by introducing functional opsin genes to restore red-green perception, but its clinical application in humans remains experimental and fraught with ethical and safety concerns (Mancuso et al., 2009). These limitations highlight the urgent need for innovative, accessible, and real-time mobile solutions that address both functional assistance and contextual adaptability for individuals living with CVD.

The Ishihara Test: Strengths, Limitations, and Implications for CVD Interventions

Figure 1

Ishihara Color Test



Note: Adapted from *OCULUS Ishihara Tests from OCULUS, Inc.* by OphthalmologyWeb n.d

The Ishihara Color Test is among the most important and reliable clinical tools for detecting red-green color vision deficiencies (Via, protanopia, red blindness, Deutanopia, green blindness), due to its simplicity, non-invasiveness, and great sensitivity, it has become widely used in clinical and community-based screenings (Hussain et al., 2021; Kumar et al., 2023). The standard 38 pseudoisochromatic plate test method allows for expedient test administration that is tailored to different severity levels, with the initial control plates evaluating baseline vision and subsequent plates progressively challenging in order to identify mild anomalous trichromacy versus total dichromacy (Cole, 2016). Recent studies affirm its diagnostic reliability against more complex methods, and clearly, the

advent of digital adaptations provides a useful method of testing utilizing smartphones (Singh & Sharma, 2020).

While the Ishihara test is quite reliable, it is also limited. It lacks sensitivity to blue-yellow deficiencies, such as tritanopia, and cannot provide a degree of color vision deficiency, which necessitates further testing (Wang et al., 2021). Nevertheless, some of its limitations are also benefits in terms of its universal applicability and cost-effectiveness, even in a low-resource context, such as the Philippines. These limitations serve as further rationale for the use of a technology-based solutions that can contribute not only to color vision differentiation, but to a more tailored, personalized, real-time, use of the testing frameworks. New mobile apps, such as ReColor, are available to augment the strengths of the Ishihara test; these apps provide adaptive remapping of color vision from respondents with an anomalous trichromacy to improve their access and living with a color vision deficiency.

Farnsworth D-15 Test in CVD Screening

Figure 2

Farnsworth D-15 Color Test Kit



Note: Adapted from Farnsworth D-15 Color Test by Ophthalmic Instruments n,d

The Farnsworth D-15 test is a widely recognized tool for diagnosing and classifying color vision deficiencies (CVD), particularly in clinical and occupational contexts. It requires individuals to arrange colored discs in perceptual sequence, effectively identifying deviations along the protan, deutan, and tritan axes (Colorlite, 2025). Its strength lies in its efficiency and high specificity, averaging around 97%, which makes it reliable for confirming moderate to severe cases of CVD (Colorlite, 2025).

Despite these advantages, the test demonstrates limited sensitivity, with detection rates ranging between 53.5% and 79.3%. This limitation reduces its effectiveness in identifying mild CVD, underscoring the need for supplementary testing when clinical precision is required (Colorlite, 2025). Nevertheless, studies such as Alotaibi et al. (2021), which applied the D-15 in screening medical students after Ishihara testing, confirm its role in efficiently detecting moderate deficiencies and informing career-related guidance.

Recent developments also highlight efforts to enhance the test’s accessibility and accuracy. Digital and technology-assisted adaptations, including electronic formats and filtered testing, are being explored to improve usability and diagnostic reach in diverse settings (Zhou et al., 2024). These innovations reinforce the continued relevance of the D-15 test while emphasizing the need for complementary

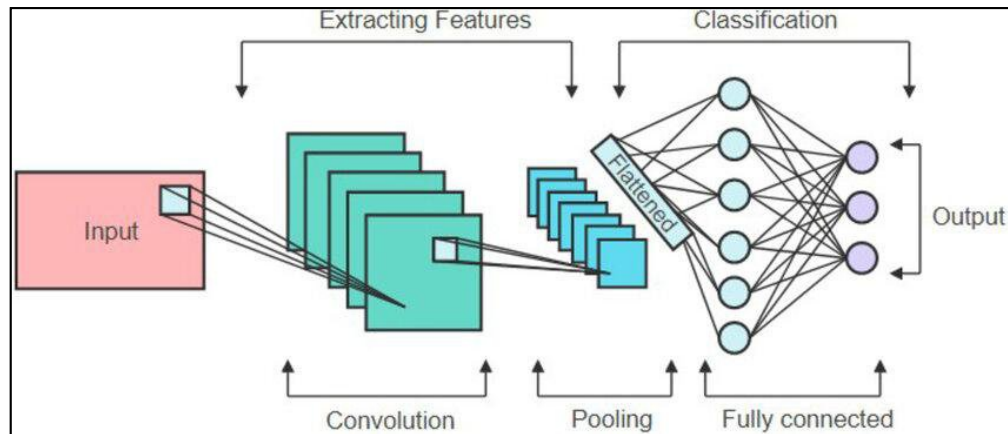
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	solutions that address its sensitivity gaps, particularly for early detection and everyday applications. Practical Applications of Machine Learning in CVD Standpoint Machine learning (ML) has emerged as a powerful approach in addressing color vision deficiency (CVD) by enabling adaptive and personalized corrections beyond the static capabilities of optical filters. Recent studies demonstrate the use of predictive models to map perceptual shifts in anomalous trichromacy and dichromacy, allowing for more precise and individualized color transformations based on real-time inputs from device cameras and user feedback (Liu et al., 2023). Deep learning methods, in particular, have shown promise in enhancing color contrast under variable lighting conditions, thereby improving usability in everyday environments compared to pre-set overlays (Chen et al., 2024). Furthermore, ML-powered augmented reality and mobile applications can generate personalized correction maps that evolve through continuous interaction with users, resulting in higher acceptance and contextual accuracy (Khan et al., 2022). Despite these advancements, challenges such as high computational demands, reliance on large training datasets, and device-specific performance constraints remain. Nonetheless, the integration of ML into assistive technologies positions it as a critical driver for next-generation, context-aware systems like ReColor, which aims to deliver low-latency, scalable, and accessible solutions for CVD populations worldwide.	
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Convolutional Neural Networks for Color Vision Deficiency Enhancement

Convolutional Neural Networks for Masking Color Vision Deficiency CNNs have completely altered the way image processing is done, allowing the features to be extracted not only automatically but also adaptively from the raw data directly. They can do this because of their specific arrangement of convolution, pooling, and fully connected layers, which enable the hierarchical learning of spatial features like edges, textures, and patterns without any human intervention (LeCun et al., 2015). The very same architecture has been responsible for placing CNNs at the forefront of object detection, segmentation, and even real-time image enhancement. They are also incredibly robust to changes in the dimensions, moving, and even distortions, which makes them very useful for real-life scenarios where lighting and background conditions are not stable (Rawat & Wang, 2017). Additionally, their integration into end-to-end frameworks increases the chances of their adaptation on mobile platforms that require fast, on-device computation. These attributes notably put CNNs at the forefront of ReColor, where adaptive feature extraction and semantic segmentation are highly rated for trichromat perception enhancement in real-time.

Figure 3

Convolutional Neural Network Architecture



Note: Adapted from An adaptive human activity recognition framework using an automated CNN architecture design by ResearchGate, 2023.

Color vision deficiency (CVD) in general, CNNs offer the possibility of personalizing and coloring the image in real time, through the analysis of the camera input, the segmentation of the areas with perceptual uncertainty, and the forecasting of the best transformations, which fit the individual needs (Khan et al., 2022). By contrast to the static filters and manual correction algorithms, the CNNs are changing along with the environment, thus providing more precise and context-sensitive color remapping (Machado et al., 2023). Nevertheless, there still are computer power training of CNNs, possible delays on low-resource devices, and the need for large labeled datasets for effective learning as hurdles. On the other hand, CNNs can be an important part of the ReColor application which is to support technology. It will be a responsible, user-centered intervention that improves color discrimination and

quality of life for individuals with anomalous trichromacy because of its adaptability and scalability

Daltonization Algorithms for Adaptive Color Remapping

Daltonization is a computational technique used to simulate and correct color vision deficiencies by shifting colors along confusion lines into more distinguishable hues, thereby enhancing perceptual differences for individuals with CVD while maintaining image naturalness (Viénot et al., 1999; Machado et al., 2009). It primarily targets red-green and blue-yellow deficiencies by compensating for impaired cone sensitivities, improving discrimination of colors that would otherwise appear indistinct

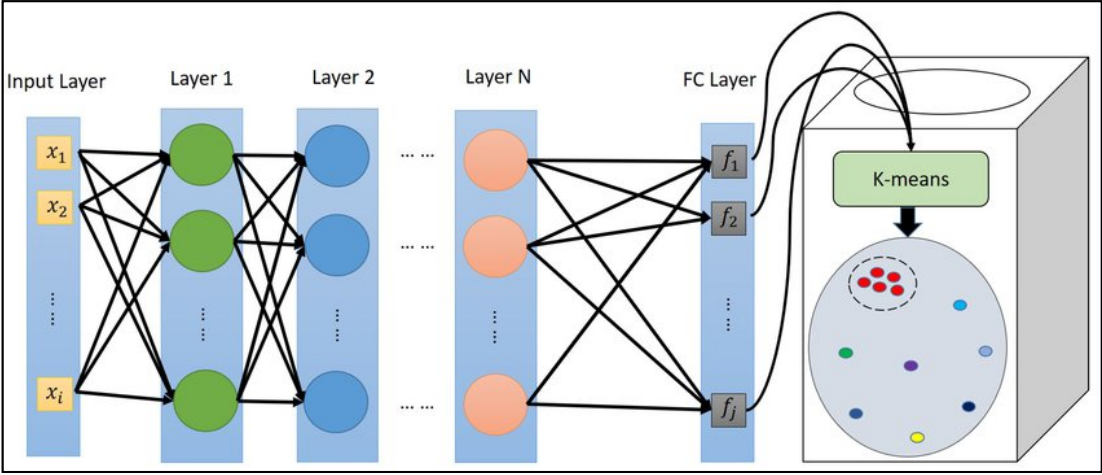
As a general rule, daltonization first bring images into a color space that is human vision-aligned such as LMS or CIE XYZ, then they apply correction matrices that redistribute the color components to the maximum possible contrast between the confusing hues (Rasche et al., 2005). The new approaches have introduced adaptive mechanisms and machine learning techniques to dynamically modify the corrections according to user profiles and environmental conditions, which in turn allows for real-time operation in mobile and augmented reality applications (Khan et al., 2022; Smith et al., 2023). There are still difficulties to be overcome like the need to find a good trade-off between enhancement and distortion, however, daltonization is still regarded as a mainstay method in CVD assistive technologies. By the combination of these algorithms, ReColor will be able to perform real-time

color remapping that considers the users' spectral deficiencies, hence, making the visual cues clearer and more accurately perceived by the users.

K-Means Clustering for Perceptual Color Segmentation

Figure 4

K-Means Clustering Architecture



Note. Adapted from The architecture of K-means clustering, by Energies, 2023

K-Means clustering is a widely used unsupervised machine learning technique for image segmentation and color quantization, where pixels are partitioned into K clusters based on feature similarity (Jain, 2010). In image processing, this allows the simplification of complex images by reducing the number of representative colors. When applied in perceptually uniform color spaces such as CIELAB or LMS, Euclidean distances align more closely with human visual perception, making K-Means particularly effective in enhancing perceptual differences (Comaniciu & Meer, 2002). For color vision deficiency (CVD),

clustering colors into distinct groups helps highlight and accentuate distinctions that may otherwise be confusing or indistinguishable.

Recent research shows that integrating K-Means into CVD enhancement pipelines improves both computational efficiency and perceptual accuracy. By remapping clusters rather than individual pixels, algorithms achieve lower latency and reduced processing overhead, which are critical for real-time mobile applications which underpins ReColor's design for efficient, mobile-based assistive vision applications. (Cheung et al., 2021).

Comparative Analysis of Local Mobile Applications for Color Vision Deficiency and ReColor

Table 3
Comparative Analysis of Local Mobile Applications for Color Vision Deficiency and ReColor

Criteria	ColorAssist (Panganiban et al., 2020)	iColorBlind (De la Cruz & Santos, 2021)	KolorAid (Mendoza et al., 2023)	ReColor (Proposed Study, 2025)
Technology Used	Android, OpenCV	Android (Java)	Android, OpenCV	React Native, TensorFlow Lite, OpenCV
Algorithms & Tools	HSV Thresholding, Simulation Models	Artificial Neural Network (ANN), Ishihara Test	Daltonization, K-Nearest Neighbor (K-NN)	K-Means Clustering, Daltonization, Convolutional Neural Networks (CNNs)
Outcome / Accuracy	97% classification accuracy for red-green deficiencies	98.95% accuracy in diagnosing CVD from Ishihara plates	Usability satisfaction: 8/10 from student participants	Targets real-time adaptive correction for anomalous trichromacy (Protanomaly & Deuteranomaly)

Local research efforts in the Philippines have explored mobile-based solutions for Color Vision Deficiency (CVD), focusing on screening, classification, and visual assistance. These projects demonstrate important progress yet also

highlight critical gaps that motivate the development of more advanced frameworks like *ReColor*.

Panganiban et al. (2020) developed **ColorAssist**, a mobile application that uses HSV thresholding and OpenCV to simulate color perception for users with CVD. The system achieved a reported accuracy of **97%** in correctly classifying red-green deficiencies, showcasing its feasibility for mobile deployment. However, the approach was limited to simulation only, without providing real-time corrective feedback or adaptive enhancements, restricting its usability in daily, practical scenarios.

Similarly, De la Cruz and Santos (2021) created **iColorBlind**, an Android-based screening tool that integrates the Ishihara test with an Artificial Neural Network (ANN) classifier for automated diagnosis. Their evaluation, based on a 50-participant dataset, yielded a 98.95% accuracy rate, confirming its diagnostic reliability. Despite this, the system was constrained to diagnosis alone, offering no corrective or assistive functionalities for users once CVD was detected.

Most recently, Mendoza et al. (2023) introduced **KolorAid**, a real-time color substitution application that implemented Daltonization and K-Nearest Neighbor (K-NN) algorithms for adaptive filtering. Usability testing with student participants produced an average satisfaction rating of **8/10**, indicating promise in enhancing visual accessibility. Nevertheless, the project's scope was limited by its restricted dataset, small-scale user testing, and device-dependent performance variability, raising questions about scalability for broader deployment.

A Hybrid Computational Framework for CVD Enhancement

Table 4

Comparative Analysis of ReColor’s Implemented Algorithms Addressing Color Vision Deficiency

Algorithm	Primary Function	Strengths	Limitations	Evaluation Metrics
K-Means Clustering	Groups pixels into clusters of similar colors to reduce computational overhead	Simplifies complex images; speeds up processing by operating on cluster centroids (Jain, 2010)	Sensitive to initialization; may oversimplify images with subtle variations (Cheung et al., 2021)	Processing speed (ms/frame), Cluster compactness (intra-cluster variance), Impact on latency
Daltonization Algorithms	Remap colors along confusion lines (LMS color space) to enhance perceptual discriminability for CVD users	Proven effectiveness in red–green and blue–yellow deficiencies; lightweight for real-time use (Viénot et al., 1999; Machado et al., 2009)	May cause unnatural color shifts; effectiveness varies by deficiency severity (Smith et al., 2023)	Color difference (ΔE , CIEDE2000), User perceptual accuracy, Image naturalness ratings
Convolutional Neural Networks (CNNs)	Extract adaptive features and learn context-specific transformations for real-time color correction	High adaptability; can personalize correction to user-specific CVD profiles (LeCun et al., 2015; Khan et al., 2022)	Computationally intensive; requires optimization for mobile (e.g., TensorFlow Lite) (Li et al., 2021)	Model accuracy (% improvement in Ishihara/D-15 test), Inference latency (ms/frame), FPS performance

he combination of Convolutional Neural Networks (CNNs), KMeans clustering, and daltonization algorithms is a great and creative way to promote color vision deficiency (CVD) correction technologies. CNNs are very powerful in extracting and segmenting features and can even adapt to the user-specific deficiency

profiles and environmental contexts to identify and prioritize the problematic color regions in an image (LeCun et al., 2015; Khan et al., 2022). K-Means clustering then takes over and carries out the task of dividing the image pixels into perceptually meaningful color groups in a uniform color space, thus shortening the images by color variance reduction and letting the color adjustment done on the cluster centroids instead of individual pixels (Jain, 2010; Cheung et al., 2021). This not only helps to cut down on the computational overhead but also keeps the important color distinctions that are relevant to the CVD users.

Daltonization algorithms then operate on these clustered colors to perform scientifically grounded color remapping along confusion lines characteristic of various CVD types, enhancing color contrast and perceptual discriminability (Viénot et al., 1999; Machado et al., 2009). This layered pipeline enables real-time, dynamic, and context-aware color correction that surpasses traditional static filters by adapting to both scene variability and individual user vision parameters.

Together, these algorithms balance computational efficiency, adaptability, and precision, constituting a scalable framework for mobile and augmented reality applications like ReColor. Nonetheless, challenges such as the high computational cost of CNN training, sensitivity of K-Means to initialization, and potential image distortion from daltonization remain areas for further refinement. Even with these limitations, the integrated system holds strong potential to significantly improve visual accessibility and quality of life for individuals with anomalous trichromacy and other color vision deficiencies across diverse settings. Such a hybrid framework underpins ReColor's design, enabling it to leverage the precision of deep learning,

the efficiency of clustering, and the accessibility of daltonization for robust mobile deployment.

TensorFlow Lite and Python for On-Device Vision Processing

Python plays a central role in artificial intelligence (AI) development due to its simplicity, readability, and extensive ecosystem of libraries that accelerate machine learning workflows (Raschka & Mirjalili, 2022). Widely adopted frameworks such as TensorFlow, PyTorch, and scikit-learn are Python-based, making it the preferred language for research, prototyping, and deployment of AI models (Abadi et al., 2016; Goyal et al., 2021). Its versatility extends from data preprocessing to deep learning model deployment, enabling rapid innovation in AI-powered solutions across diverse domains, including healthcare, vision enhancement, and accessibility technologies.

Within this ecosystem, TensorFlow Lite serves as a lightweight extension optimized for mobile and embedded platforms, supporting real-time inference with low latency and efficient resource usage (TensorFlow, 2024). Combined with Python’s development advantages, it enables streamlined model conversion and deployment for convolutional neural networks (CNNs) in adaptive image-processing tasks, such as real-time color vision deficiency correction (Adyani et al., 2025; Khan, Li, & Chen, 2022). While limitations exist, such as reduced flexibility for complex architectures and performance variability across devices (Li, Chen, & Patel, 2021), Python’s integration with TensorFlow Lite ensures scalable, efficient, and accessible AI development. For ReColor, this synergy allows adaptive vision enhancement models to run effectively on-device, maintaining responsiveness while safeguarding user experience.

Firestore as a Backend Solution for User Profile and Test Result Management in ReColor

The rise of mobile health applications highlights the importance of secure and scalable backend systems. Firestore, Google’s Backend-as-a-Service (BaaS), offers authentication, real-time databases, and offline persistence, making it well-suited for assistive tools like ReColor. Through Firestore Authentication, the app enables secure user onboarding, while Firestore’s flexible NoSQL database supports dynamic updates of user profiles, including customized color correction settings. Its offline-first capability is especially critical in the Philippine context, where unstable connectivity can limit consistent access (Google, 2025; Tiwari & Agrawal, 2021).

In addition to user profile management, Firestore facilitates secure storage of diagnostic results such as Ishihara tests. Firestore’s timestamped, document-based structure allows longitudinal monitoring of users, while cloud functions support anonymization for safe clinical collaboration with the Philippine Eye Research Institute. Its compatibility with React Native enables smooth cross-platform deployment, while Firestore Cloud Messaging enhances engagement through awareness and career guidance notifications. These combined features provide ReColor with a reliable, adaptive, and scalable infrastructure for delivering accessible assistive technology tailored to Filipino users with color vision deficiency (Gunjal et al., 2024).

React Native and OpenCV in Mobile-Based Real-Time Image Processing

React Native is a widely adopted framework for cross-platform mobile development, allowing developers to build native-like applications using JavaScript and React (Moshkovich et al., 2021). Its cross-platform performance makes it suitable for real-time, computation-heavy tasks such as adaptive color vision enhancement. Extensions strengthen its capabilities: React Native Vision Camera provides high-performance access to native camera APIs for low-latency video streaming (Hahn et al., 2022); React Native Reanimated delivers GPU-accelerated animations and gesture handling for responsive user interfaces in health-related applications (Zhao et al., 2020); and React Native FS supports asynchronous read/write operations for efficient on-device caching of images and logs (Nguyen & Tran, 2023).

Complementing these, OpenCV is a robust open-source computer vision library widely applied in mobile applications for real-time image analysis and manipulation. Its optimized native implementations enable efficient execution of tasks such as color space transformations, K-Means clustering, and filtering, which are essential for adaptive color remapping in users with color vision deficiencies (Bradski & Kaehler, 2008; Sharma et al., 2021). The integration of React Native with OpenCV, often through native module bridging, ensures high-performance assistive vision applications by combining seamless user interfaces with powerful image processing capabilities (Kim et al., 2022). The integration of React Native and OpenCV supports *ReColor's* vision of a cross-platform, assistive mobile application, combining a seamless user interface with optimized image processing capabilities.

Android Core App Quality Standards as a Benchmark for Evaluating ReColor

The Android Core App Quality Guidelines, established by Google, serve as a fundamental benchmark for ensuring usability, reliability, and performance in mobile applications. These guidelines emphasize consistent UI/UX patterns, adherence to Material Design principles, and seamless navigation to provide a predictable and user-friendly experience (Android Developers, 2023). Empirical studies demonstrate that compliance with platform-specific standards improves app learnability and user satisfaction by leveraging familiar interaction patterns (Moshkovich et al., 2021). For assistive tools such as ReColor, strict adherence to these principles is essential to guarantee accessibility for individuals with Color Vision Deficiency (CVD), particularly in resource-constrained contexts like the Philippines where Android dominates the mobile market with over 90% share (Statcounter, 2024).

Beyond usability, Android’s evaluation framework highlights key technical dimensions such as performance, stability, and security. Google’s Android Vitals program encourages developers to monitor metrics including crash rates, application not responding (ANR) errors, and battery usage, which directly impact user retention and satisfaction (Burke et al., 2022). The guidelines also mandate minimal permission requests and robust privacy practices to maintain user trust and comply with security standards (Luo et al., 2019). Complementary to these platform-specific measures, accessibility principles outlined in WCAG 2.2 reinforce inclusive practices such as sufficient contrast ratios, descriptive labels, and non-color cues for

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	equitable use (W3C, 2023). Collectively, these standards create a holistic framework that parallels international models like ISO/IEC 25010 while tailoring them to Android’s ecosystem. Applied to ReColor, Android Core evaluation ensures the app not only functions effectively but also meets expectations for responsiveness, inclusivity, and trustworthiness, thus supporting its role as a scalable vision-assistive tool. Accessibility Standards in Software Development: WCAG 2.2 The Web Content Accessibility Guidelines (WCAG) 2.2, developed by the World Wide Web Consortium (W3C), provide internationally recognized standards for ensuring digital accessibility across diverse disabilities (W3C, 2024). Building on previous versions, WCAG 2.2 emphasizes four principles perceivable, operable, understandable, and robust while introducing new criteria such as enhanced focus visibility, accessible authentication, and target size requirements to improve usability for individuals with low vision, limited motor abilities, and cognitive impairments (Axelerant, 2023; Digital NSW, 2023). Increasingly adopted as a benchmark for accessibility compliance, WCAG 2.2 supports legal and regulatory mandates while guiding both automated and manual evaluation methods (Knowbility, 2022; GOV.UK, 2024). For applications like <i>ReColor</i>, these guidelines ensure that accessibility is systematically integrated, enabling inclusive design for users with color vision deficiencies and aligning the system with global accessibility standards (Peimbert-García et al., 2023).	
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	Best Practices in Colorblind-Friendly Palette Design and Their Application in ReColor.	
	Best practices for developing colorblind-friendly palettes emphasize pre-planning color schemes, avoiding problematic color pairs (e.g., red/green, green/brown, blue/gray), and maintaining accessible contrast ratios as defined by the Web Content Accessibility Guidelines (WCAG) 2.2, which recommend a minimum contrast ratio of 4.5:1 for text and critical visuals (Smith et al., 2022; W3C, 2023). Integrating non-color cues such as textures, patterns, or symbols further enhances clarity for individuals with severe deficiencies by reducing reliance on color alone (Chen & Zhao, 2023). Tools such as Coblis simulators and palette generators enable designers to validate accessible color schemes efficiently, ensuring usability across digital platforms (Patel et al., 2020; Venngage, n.d.). In the context of ReColor, these practices provide design guidelines that directly inform how adaptive daltonization and clustering algorithms remap colors into more distinguishable palettes without compromising usability. Empirical studies confirm that implementing colorblind-friendly palettes improves comprehension and reduces misinterpretation in data visualization and user interfaces, directly enhancing user engagement and decision-making (Murray et al., 2021). Their adoption is particularly critical in domains such as healthcare, education, and public communication, where accurate interpretation of visual data is essential. For ReColor, embedding these accessibility practices strengthens the app’s capability to generate color-corrected visuals that are both perceptually distinct and	
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compliant with global accessibility standards. This alignment ensures the system not only enhances real-time vision for anomalous trichromats but also upholds principles of inclusivity and equity in mobile assistive technology design (Johnson & Roberts, 2020; Garcia & Lee, 2023).

Synthesis

The reviewed literature highlights persistent challenges and evolving solutions in addressing color vision deficiency. Classical diagnostic methods such as the Ishihara plates and the Farnsworth D-15 test remain standard tools for detection, but they are limited in sensitivity for mild anomalous trichromacy and do not provide functional assistance beyond diagnosis (Kumar et al., 2023; Singh & Sharma, 2020; Wang et al., 2021). Philippine prevalence studies report that 5.17 percent of male high school students in Metro Manila are affected, predominantly by the deutan type (HERDIN, n.d.; Cruz et al., 2010). These findings emphasize the need for both improved early detection and post-diagnostic support that are tailored to local contexts and resource constraints.

Existing assistive technologies vary in cost, scope, and effectiveness. Optical aids such as spectral filtering glasses can provide immediate subjective improvement but are constrained by lighting conditions and high cost (Almutairi & Alharbi, 2022; Pattie et al., 2022). Software-based solutions are more affordable and accessible but frequently lack real-time adaptability and generalizability outside of screen-based contexts (Shinohara & Wobbrock, 2023; Machado et al., 2023). Experimental therapies such as gene transfer show biological promise in animal models but remain ethically and technically premature for routine human use (Mancuso et al., 2009).

Local computational and student-led projects demonstrate grassroots innovation but reveal gaps in robustness and scalability. Examples include ColorAssist, which used RGB filtering and thresholding (Panganiban et al., 2020), iColorBlind, which implemented HSV manipulation with OpenCV (De la Cruz & Santos, 2021), and KolorAid, which applied deep learning segmentation techniques (Mendoza et al., 2023). While these projects validate the feasibility of mobile CVD aids in the Philippine setting, they generally lack hybrid algorithm integration, comprehensive evaluation, and adaptation to varied device capabilities.

ReColor is positioned to address these gaps by combining established and complementary methods into a hybrid pipeline that integrates K-Means clustering for efficient color grouping, daltonization for perceptual remapping, and convolutional neural networks for adaptive, user-specific transformations. This integrated approach aims to deliver real-time, on-device enhancement that balances perceptual accuracy and computational efficiency. The adoption of Android Core App quality guidelines as the evaluation benchmark ensures that technical assessment focuses on mobile-relevant attributes such as performance, stability, accessibility, and compatibility (Android Developers, 2024).

Finally, the literature and policy context point to an unmet need for tools that bridge detection, functional assistance, and public health objectives. Republic Act No. 11358 mandates early vision screening for young children but does not explicitly include CVD detection or sustained assistive measures (LawPhil, 2019). By embedding Ishihara-based screening, adaptive correction, and awareness materials

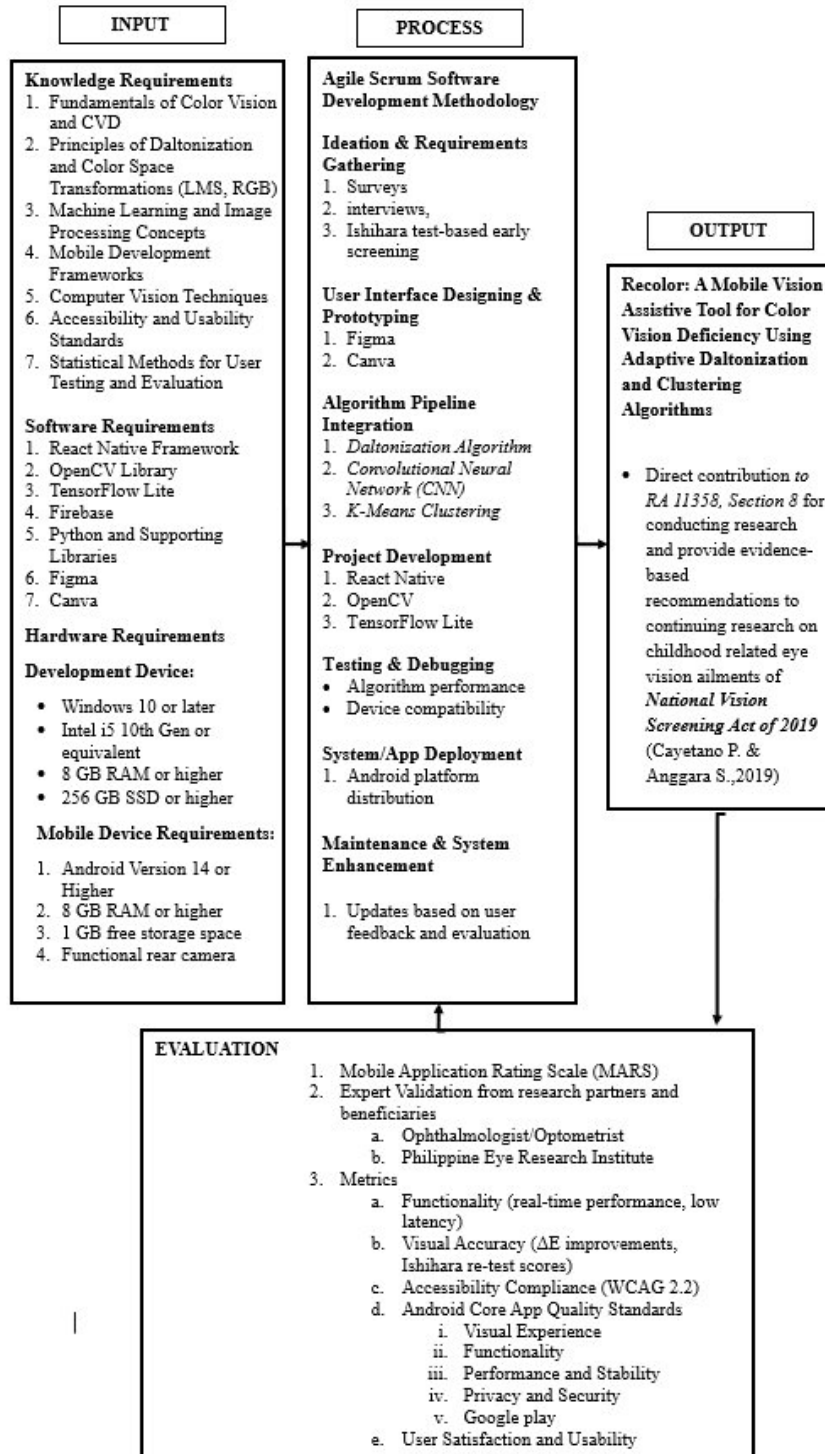
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for career guidance, ReColor directly supports the aims of national screening initiatives while extending their practical utility for daily living and long-term inclusion.

Conceptual Framework

Figure 5

Conceptual Framework of ReColor



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	The conceptual framework of this study is structured around the Input-Process-Output (IPO) model to ensure a systematic integration of technical, user, and policy requirements. The Input consists of four major components: (1) Knowledge requirements, which include theoretical foundations on Color Vision Deficiency (CVD), the principles of Daltonization and color space transformations (LMS, RGB), machine learning and image processing concepts, mobile development frameworks, computer vision techniques, accessibility and usability standards such as WCAG 2.2, and statistical methods for user testing and evaluation; (2) Software requirements, including React Native for cross-platform deployment, OpenCV for image processing, TensorFlow Lite for on-device machine learning, and Firebase for authentication, user profile management, test result storage, and cloud functions; (3) Hardware requirements, such as development devices with sufficient computational capacity (Intel i5 10th Gen or higher, 8 GB RAM, 256 GB SSD) and mobile devices meeting Android version 14 or higher, with a functional rear camera for real-time testing; and (4) Data requirements, which involve Ishihara test plates, benchmark image datasets, and user feedback gathered from CVD and non-CVD individuals. The Process follows a Modified Agile–Spiral methodology. It begins with ideation and requirements gathering through surveys, interviews, and Ishihara-based early screening. User interface design and prototyping are carried out with Figma and Canva, while the algorithmic pipeline integrates Daltonization for perceptual remapping, K-Means clustering for efficient pixel grouping, and Convolutional	
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Neural Networks (CNNs) for adaptive user-specific transformations. Project development involves React Native, TensorFlow Lite, and OpenCV integration, with Firebase managing authentication, real-time data synchronization, and cloud-based notifications. The testing and debugging phase ensures algorithm accuracy and device compatibility, followed by deployment on the Android platform via Google Play Store. Maintenance and system enhancement are achieved through iteration cycles, guided by weekly sprint meetings and bi-monthly consultations with the Philippine Eye Research Institute (PERI) for clinical validation.

The Output is the fully developed mobile application, ReColor: A Mobile Vision Assistive Tool for Color Vision Deficiency Using Adaptive Daltonization and Clustering Algorithms, which provides both screening (via embedded Ishihara testing) and enhancement (via real-time adaptive color correction) functionalities. The system directly contributes to Republic Act No. 11358 (National Vision Screening Act of 2019) by supporting early detection, advancing research, and delivering evidence-based interventions to improve the daily lives of Filipinos with CVD.

The Evaluation of ReColor is conducted through both quantitative and qualitative approaches. Quantitatively, evaluation metrics include functionality (real-time performance, latency) and visual accuracy (ΔE improvements, Ishihara re-test scores). Qualitatively, usability and accessibility are assessed using the Mobile Application Rating Scale (MARS) and Android Core App Quality Guidelines under the scope of Visual Experience, Functionality, Performance &

Stability, Privacy and Security and Google Play Guidelines. Expert validation is obtained through structured evaluations from ophthalmologists, optometrists, IT specialists, and research partners at the Philippine Eye Research Institute. Together, these measures ensure that the system not only demonstrates technical feasibility but also meets user-centered and policy-driven objectives.

Definition of Terms

For the readers to fully comprehend this study, the following are the notable terms that have been utilized in the study:

Adaptive Color Remapping: A process of adjusting and transforming colors in images or videos so that they become more distinguishable for people with color vision deficiencies.

Anomalous Trichromacy: A type of color vision deficiency where all three types of cone cells in the eye are present, but one type perceives colors differently than normal, leading to difficulty distinguishing certain shades (e.g., red or green).

CIELAB (Lab) Color Space: A color space designed so equal distances roughly match human-perceived differences; commonly used in color difference calculations.

CIE XYZ Color Space: A device-independent color space that models human vision and is often used as an intermediate space in color transformations.

Clustering Inertia: In K-Means clustering, the sum of squared distances of points to their cluster centers, representing cluster compactness.

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Clustering Purity: A metric that measures how well each cluster corresponds to a single ground-truth category, often used to evaluate segmentation quality.

Color Vision Deficiency (CVD): A condition that impairs the ability to distinguish certain colors, commonly referred to as color blindness.

Confusion Lines: Lines in a color space along which colors appear identical to a person with a specific type of CVD, used in daltonization techniques.

Convolutional Neural Network (CNN): A type of deep learning model designed to process and analyze images by automatically learning to detect visual features such as edges, shapes, and colors.

Cross-Device Benchmarking: The process of testing application performance and quality across multiple smartphone models to evaluate consistency.

Cross-Platform Development: Building applications that can run on multiple operating systems (e.g., Android and iOS) using a single codebase.

Daltonization: An image processing technique used to adjust colors along confusion lines so that individuals with color blindness can better distinguish them.

Deep Learning: A branch of artificial intelligence (AI) that uses multi-layered neural networks to learn complex patterns from data.

Deuteranomaly: A type of anomalous trichromacy where sensitivity to green light is reduced, making it difficult to distinguish green-related shades.

Farnsworth D-15 Test: A clinical color vision test where users arrange colored caps in sequence, helping classify moderate to severe color vision deficiencies.

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Figma: Figma is a cloud-based UI/UX design platform that was first launched in September 2016. It is primarily used for creating wireframes and prototypes for websites and mobile applications.

GPU (Graphics Processing Unit): A specialized processor originally designed for rendering graphics, now widely used to accelerate artificial intelligence and image processing tasks.

Hardware Acceleration: The use of specialized hardware (such as GPUs and NPUs) to perform computations faster than on general-purpose CPUs.

Image Processing: The manipulation and analysis of digital images to improve quality, extract information, or adapt them for specific purposes.

Ishihara Test: A widely used color vision test that uses pseudoisochromatic plates with colored dots forming numbers, primarily detecting red-green deficiencies.

ISO/IEC 25010: An international software quality standard that defines characteristics such as functionality, performance, usability, reliability, and maintainability.

K-Means Clustering: A machine learning algorithm that groups data points, such as pixels, into K clusters based on similarity, often used in image segmentation.

LMS Color Space: A color space based on human cone responses (Long, Medium, Short), often used in modeling and simulating CVD.

Low Latency: A very short delay between input and output, crucial for real-time applications such as live video enhancement.

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	Machine Learning: A field of computer science that enables systems to learn patterns from data and improve performance on tasks without explicit programming. Neural Processing Unit (NPU): A specialized chip in modern smartphones designed specifically to accelerate artificial intelligence tasks such as real-time video processing. On-Device Inference: Running artificial intelligence models directly on a smartphone or device without relying on cloud servers, ensuring privacy and responsiveness. OpenCV (Open Source Computer Vision Library): A widely used open-source library that provides tools for real-time image and video analysis. Paired T-Test. A paired samples t-test is another type of t-test used when comparing scores measured or gathered from the same group. Processing Latency (ms/frame): The amount of time required for a system to process a single video frame, measured in milliseconds. Protanomaly: A type of anomalous trichromacy where sensitivity to red light is reduced, causing difficulty distinguishing red-related shades. Pseudoisochromatic plates: A series of plates used to assess color vision, featuring dotted numbers of one color against a contrasting background. Quantization: A technique in deep learning that reduces model size and speeds up inference by representing numbers with fewer bits, sometimes at the cost of accuracy.	
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RA 11358: National Vision Screening Act mandates the establishment of the National Vision Screening Program (NVSP) for kindergarten pupils.

React Native: A cross-platform framework that enables mobile app development for both iOS and Android using JavaScript and React.

Real-Time Processing: The capability of a system to analyze and respond to input data instantly or with minimal delay.

Semantic Segmentation: A computer vision method that assigns a class label to each pixel in an image, helping identify objects and regions of interest.

SQLite: A lightweight, file-based database engine used to store and manage data locally within mobile applications.

SSIM (Structural Similarity Index): An image quality metric that measures the similarity between two images based on structural information, luminance, and contrast.

TensorFlow Lite (TFLite): A lightweight version of TensorFlow designed to run machine learning models efficiently on mobile and embedded devices.

User Interface (UI): The visual and interactive elements of an application through which users engage with the system.

Vision Enhancement: The process of using technology to improve how people see, particularly for those with visual impairments or color vision deficiencies.

WCAG 2.2 (Web Content Accessibility Guidelines): An international standard ensuring digital content is accessible to people with disabilities, focusing on perceivable, operable, understandable, and robust design principles.

CHAPTER III

METHODOLOGY

Research Design

The study implemented a mixed methods design to evaluate ReColor, a mobile vision assistive tool for individuals with color vision deficiency. The design combined qualitative inquiry and quantitative evaluation to examine the system's hybrid algorithmic pipeline, usability, accessibility, and impact on awareness, daily functioning, and career guidance. The Philippine Eye Research Institute served as the primary clinical partner for expert validation and pilot recruitment.

For the qualitative component, data were collected through semi-structured interviews, focus group discussions, and open-ended survey items with purposively sampled end users (clinically diagnosed or self-reported protanomaly and deuteranomaly) and non-CVD participants. In addition, expert evaluators including ophthalmologists, optometrists, and software specialists provided professional feedback through structured interviews and validation checklists. Qualitative data will be transcribed, coded, and analyzed using thematic analysis. Trustworthiness will be enhanced through investigator triangulation, member checking, and an audit trail.

For the quantitative component, the study measured diagnostic validity, algorithmic performance, usability, and system responsiveness. The adaptive color correction pipeline integrating K-Means clustering, daltonization, and convolutional neural networks was evaluated using domain-appropriate metrics such as CIEDE2000 ΔE and SSIM for perceptual accuracy, precision, recall, F1-score and confusion matrices for the CNN color identifier, and sensitivity, specificity, accuracy, and Cohen's kappa for the embedded Ishihara screening. Usability and accessibility were assessed with Android Core App quality benchmarks and standardized questionnaires. Statistical analysis will include descriptive statistics, reliability testing (Cronbach's alpha), and inferential tests such as paired t-tests or Wilcoxon signed-rank tests for pre-post comparisons, with effect sizes and 95 percent confidence intervals reported.

A prospective pilot cohort provided within-subject, repeated-measures data on feasibility and preliminary efficacy. PERI advised a minimum clinical cohort of five CVD participants for pilot testing; the cohort will be supplemented by additional non-CVD users and technical evaluators for broader usability assessment. Each cohort member will follow a standardized protocol including baseline screening, controlled and real-world trials across lighting conditions, and post-trial assessments. Pilot data will inform power calculations for future studies and guide iterative refinement of both algorithms and user interfaces.

Dataset Description

a. Screening dataset.

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Responses and recorded outputs from the embedded digital Ishihara plates collected during user testing. These data support evaluation of the screening module and calculation of diagnostic metrics such as accuracy, sensitivity, specificity, and Cohen's kappa.

b. *Real-world scene captures.*

Photographic frames captured from participants using target devices in diverse environments (indoor, outdoor, varied lighting) to train and evaluate the color correction and CNN-based color identifier modules. These images represent everyday scenes, user interfaces, product packaging, clothing, and traffic/wayfinding cues that are relevant to color-discriminative tasks.

c. *Color patch and calibration dataset.*

Images of standardized color targets and charts captured on each device used for development and testing (for example, X-Rite-style color checkers) to derive device-specific color calibration transforms and to compute colorimetric ground truth.

d. *Annotated color patch dataset.*

Manually labeled color patch crops or cluster centroids used to train the CNN color identifier and to map cluster centroids to semantic color labels.

e. *Synthetic and simulated CVD dataset.*

Images produced by applying physiologically based CVD simulation matrices at multiple severity levels to augment training data and to test robustness of daltonization and hybrid algorithms.

Algorithms and Model Training

Overview

The algorithmic foundation of ReColor integrates a hybrid adaptive vision enhancement pipeline designed to identify, analyze, and correct color distortions experienced by individuals with Color Vision Deficiency (CVD). This section presents the systematic process of algorithm development and model training that underpins the ReColor system. The model is composed of three primary computational modules:

- (A) *K-Means Clustering* for color segmentation and pixel grouping,
- (B) a *Convolutional Neural Network (CNN)* for pattern recognition and color transformation mapping, and
- (C) a *Daltonization Algorithm* for perceptual color remapping based on simulated CVD responses.

Each module was developed and trained independently to ensure optimal parameterization and model performance before integration into the unified hybrid pipeline. The K-Means algorithm provides an efficient preprocessing mechanism that reduces image complexity by segmenting pixel groups based on color similarity. The CNN model serves as the system’s intelligent core, learning to interpret and transform color distributions according to the user’s specific deficiency profile. Finally, the Daltonization algorithm applies color correction matrices that enhance perceptual distinction between previously indistinguishable color regions while preserving luminance and overall image fidelity.

The overall training and development process followed a sequential and modular approach to ensure both computational efficiency and real-time compatibility. Each algorithm underwent an iterative training cycle involving hyperparameter tuning, validation, and performance testing using both quantitative metrics and qualitative feedback. Model optimization, conversion, and deployment were performed through TensorFlow Lite to ensure responsiveness and low-latency performance on mobile platforms.

A) K-Means Clustering Module

The K-Means Clustering Module served as the preliminary stage of the ReColor system’s hybrid processing pipeline. Its primary function was to segment input images into perceptually distinct color regions, thereby simplifying subsequent computation in the Convolutional Neural Network (CNN)-based color identifier and Daltonization modules. By grouping pixels into representative color families, the system effectively reduced image complexity while preserving essential chromatic information.

Dataset Preparation

The dataset for the K-Means clustering module was constructed using both custom-curated image samples and standardized color definitions sourced from Python’s built-in libraries, *webcolors* and *colors*. These libraries provide a comprehensive mapping of recognized color names to their corresponding RGB and HEX values, ensuring color fidelity and reproducibility throughout the training

process. The dataset primarily focused on three core color families red, green, and blue serving as the foundational categories for color segmentation and classification.

Ttry (*Pillow*). Each image was resized to a standard dimension of 64×64 pixels and converted into the *RGB color space* to maintain uniformity. The pixel data were extracted and reshaped into flattened numerical arrays using *NumPy*, forming the input matrix for clustering. These arrays allowed the K-Means algorithm to identify and group pixels into distinct clusters based on their chromatic similarity.

For the initial experiment, the clustering parameter was set to $n = 3$, corresponding to the three primary color families. In later iterations, the number of clusters was expanded to $n = 9$ to represent a more comprehensive range of perceptual colors: *red, orange, yellow, green, cyan, blue, purple, white, and black*. This progression allowed the model to simulate a broader spectrum of color segmentation, which is crucial for adaptive color transformation in users with Color Vision Deficiency (CVD).

Table 5

K-Means Clustering Color Dataset Sources and Configuration

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Dataset Component	Source/Library	Color Space Used	Purpose
Primary Color Families	Custom dataset (Red, Green, Blue)	n = 8 clusters	Defines foundational training categories for clustering
Reference Color Values	webcolors (Python Library)	RGB	Provides standardized name-to-RGB color mappings
Supplementary Palette	colors (Python Package)	HSV	Expands classification spectrum for finer clustering
Image Preprocessing	Pillow (PIL)	RGB	Converts and resizes image samples for uniform input
Data Handling and Processing	NumPy	RGB Arrays	Flattens image data for algorithmic computation

The resulting dataset establishes the foundation for subsequent model development in the Convolutional Neural Network (CNN) and Daltonization modules, maintaining chromatic consistency throughout the ReColor pipeline.

Algorithm Configuration and Training

The KMeans class from scikit-learn's library was employed by the algorithm to conduct unsupervised color clustering through the process of minimizing the Euclidean distance. A major color of the image was represented by each cluster centroid, thus, the major colors were captured by the algorithm while the variations were filtered out. The first tests used $k = 3$ clusters to assess the primary RGB families' color separation at the basic level. Once confirmed, the number of clusters was increased to $k = 9$, which matched the perceptual color categories Red, Orange, Yellow, Green, Cyan, Blue, Purple, White, and Black.

This expansion improved the model’s color granularity and perceptual differentiation, ensuring that each cluster corresponded more closely to naturally distinguishable hues a crucial factor in aiding users with color vision deficiency.

Cluster Mapping and Visualization

Figure 9 illustrates an example output using $n = 3$ clusters, demonstrating the algorithm’s ability to segment an image into distinct color regions. Future implementations while

Figure 10 with $n = 9$ clusters provided finer segmentation and enhanced accuracy in identifying and remapping visually confusing colors for CVD users.

Table 7 presents the extended mapping of the nine color clusters, defining each centroid as a perceptual representative of the corresponding color family. These values will be dynamically updated during iterative retraining to reflect improved chromatic precision.

Figure 6

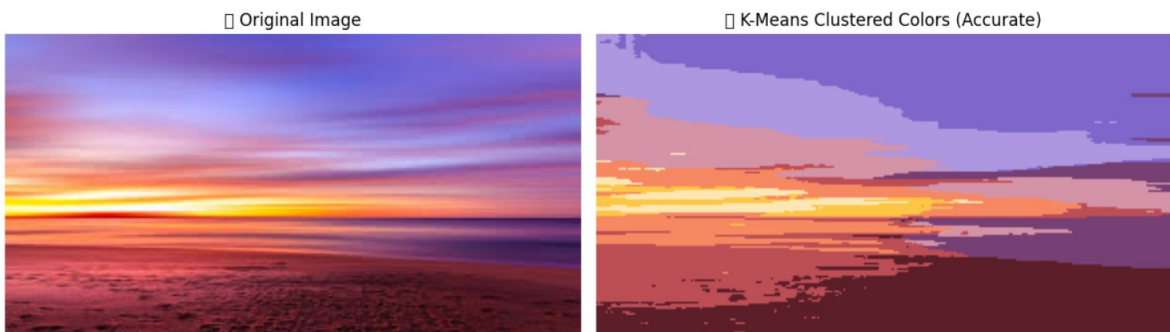
Color Segmentation Output Using K-Means Clustering ($n = 3$)



Note: Figure 9 should display the original image on the left and the clustered image on the right, following the matplotlib output.

Figure 7

Color Segmentation Output Using K-Means Clustering ($n = 9$)



Note: Figure 10 should display the original image on the left and the improved clustered image on the right, following the matplotlib output.

Table 6

K-Means Cluster Mapping for $n = 9$

Cluster	Assigned Label	Representative RGB Centroid
1	Red	(255, 0, 0)
2	Orange	(255, 165, 0)
3	Yellow	(255, 255, 0)
4	Green	(0, 255, 0)
5	Cyan	(0, 255, 255)
6	Blue	(0, 0, 255)
7	Purple	(128, 0, 128)
8	White	(128, 0, 128)
9	Black	(128, 0, 128)

2) CNN-Based Color Identifier Module

Model Objective and Role in Their Classification

The CNN Color Identifier module, based on Convolutional Neural Networks (CNN), is responsible for the automatic classification of raw RGB image inputs into a comprehensive 19-class color taxonomy consisting of: *beige/tan, black, blue, brown, cyan, gold, gray, green, lime, magenta, navy, orange, pink, purple, red, silver, teal, white, and yellow.*

CNN performs three essential roles: (1) standardizing color representations for uniform processing in downstream modules, (2) supporting analytical validation through color frequency and misclassification metrics, and (3) enabling lightweight, real-time inference through TensorFlow Lite deployment for mobile and edge devices.

The model adopts an EfficientNetB0 backbone, optimized for parameter efficiency and generalization. This architecture provides reliable color differentiation with minimal computational cost, strengthening ReColor’s ability to enhance visual accessibility for users with Color Vision Deficiency (CVD).

Dataset Composition and Preprocessing

The dataset for the CNN Color Identifier was programmatically generated using the Python libraries webcolors and colors, which enumerate canonical color names and derive representative RGB triplets. To support the expanded taxonomy, specifically metallic classes (silver, gold) and textured neutrals (beige/tan, brown), the generation pipeline included simulation of multi-band highlights and specular reflections. Synthetic image tiles and composite scenes were created to simulate real-world variation through subtle gradients, contrast differences, and mild textures, ensuring that the model learned perceptual color boundaries rather than trivial solid color blocks.

Table 7

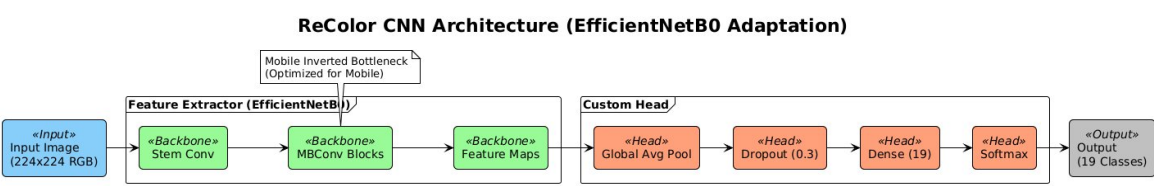
CNN Dataset Composition and Preprocessing Summary

Aspect	Description
Image Size	Resized to 224×224 pixels using bilinear interpolation; center cropping applied to maintain aspect ratio.
Normalization	Applied via <code>tf.keras.applications.efficientnet.preprocess_input</code> to align pixel statistics with pretrained EfficientNet parameters.
Label Encoding	One-hot encoding of 19 discrete classes ; optional label smoothing applied to calibrate output confidence.
Augmentation	Advanced pipeline: Random horizontal flips, rotations ($\pm 5\%$), zoom ($\pm 10\%$), contrast jitter, ISONoise (sensor grain simulation), and JPEG compression artifacts.
Hue Perturbations	Conditional hue/saturation shifts enabled in extended training mode to enrich inter-class variance, particularly for adjacent hues (e.g., navy vs. blue).

Network Architecture and Design

Figure 8

ReColor CNN Architecture Using EfficientNet80 Adaptation



CNN employs *transfer-learning architecture*, using a pretrained *EfficientNet-B0* feature extractor followed by a compact, custom classification head.

Table 8

CNN Core architectural components

Component	Description
Convolutional Stem & MBConv Blocks	Depthwise and pointwise convolutions with squeeze-and-excitation layers that progressively expand feature channels while downsampling spatial dimensions.
Activation Function	Swish (EfficientNet default) for non-linearity; final dense layer uses softmax for multiclass probability outputs.
Pooling Mechanism	Global Average Pooling converts spatial feature maps into fixed-length vectors to enhance stability under geometric augmentations.
Regularization	Dropout (rate = 0.3), label smoothing ($\epsilon = 0.1$), and AdamW weight decay ($1e-4$) to mitigate overfitting.
Output Layer	Dense (19, softmax) trained using categorical cross-entropy loss.

Table 9

CNN Training Configuration Summary

Parameter	Configuration
Epochs	Multi-stage: Baseline training (8 epochs) followed by targeted fine-tuning epochs.
Optimizer	AdamW (fallback to Adam if unsupported)
Learning Rate	$3e-4$ cosine decay with restart; ReduceLROnPlateau backup
Strategies	Class Weighting: Escalated weights (e.g., 15.0) applied to hard-negative classes (Navy) during fine-tuning.

Framework

TensorFlow 2.20.0 + Keras 3.12.0

Dataset Preparation and Preprocessing

The dataset used for training the CNN Color Identifier was programmatically generated to accommodate the expanded 19-class taxonomy. While libraries like webcolors provided the base RGB triplets, significant data engineering was required to synthesize realistic representations for new classes such as silver and gold (simulating metallic sheen) and to consolidate redundant classes (merging 'dark red' into 'red' to balance the distribution). A total of approximately 7,600 synthetic image samples (400 per class) were generated, equally distributed to ensure balanced training.

Synthetic image tiles were rendered to include contrast variations, mild textures, and gradient transitions. This approach addressed the lack of publicly available labeled datasets for fine-grained color identification while maintaining full control over lighting and saturation parameters.

Preprocessing and normalization were applied consistently across all images. Each input was resized to 224×224 pixels using bilinear interpolation. Pixel values were scaled and normalized through *tf.keras.applications.efficientnet.preprocess_input*. Labels were one-hot encoded, with optional label smoothing ($\epsilon = 0.1$).

To bridge the gap between synthetic training data and real-world camera feeds, a robust augmentation strategy was implemented via the Albumentations library. Transformations included geometric shifts (rotation, zoom) and, crucially,

photorealistic noise injections (ISONoise, Blur, Compression) to simulate low-quality camera sensors. Furthermore, class-conditional augmentations (e.g., targeted darkening and saturation shifts) were applied to specific classes like Navy and Blue to force the model to learn distinct decision boundaries between spectrally similar colors.

Table 10

CNN Summary of Dataset Composition and Preprocessing Parameters

Aspect	Description
Total samples	7,600 images (Balanced 19-class distribution; 400 per class)
Image size	224 × 224 pixels
Preprocessing	Bilinear resize, normalization (EfficientNet standard), one-hot encoding
Augmentations	Geometric (Flip/Rotate/Zoom), Photorealistic (ISONoise/JPEG/Blur) , Class-conditional hue shifts
Label smoothing	0.1 (optional)
Dataset split	Stratified Split: 340 Training / 60 Validation per class

Figure 9

CNN Sample Augmented Color Dataset



Note: Figure 11 illustrates representative samples from the augmented dataset, highlighting hue shifts, illumination variance, and perceptual boundary diversity.

Training and Evaluation Metrics

The CNN Color Identifier module underwent a rigorous multi-stage training process to ensure robustness across all 19 color categories. Metrics including loss, accuracy, and macro-F1 score were monitored per epoch.

Initial baseline training (8 epochs) identified specific confusion clusters, notably a directional collapse between Navy and Blue. To resolve this, a targeted fine-tuning strategy was implemented using Class Weighting. The model was fine-tuned with significantly higher penalties (Weight = 15.0) for misclassifying Navy samples. This strategy successfully differentiated the classes without degrading overall accuracy.

The final model achieved a Macro-F1 score of 0.99 on the validation set. Early stopping was employed based on the stabilization of the validation macro-F1 score, and the best-performing weights were preserved.

Table 11

Training Configuration and Parameters for CNN Color Identifier Module

Parameter	Configuration
Optimizer	AdamW (weight decay = 1e-4)
Learning Rate	3×10^{-4} with cosine decay restarts
Epochs	8 Baseline + Targeted Fine-tuning steps
Batch Size	32
Input Size	224×224×3
Loss Function	Categorical Cross-Entropy ($\epsilon = 0.1$ label smoothing)
Regularization	Dropout (0.3), Weight Decay (1e-4), Class Weights (High penalty for Navy/Blue separation)
Metrics	Accuracy, Macro-F1 (Primary), Precision, Recall

3) Daltonization Algorithm

This section describes the theoretical background, the mathematical models, and the specific architectural choices that have been made in the application ReColor to achieve robust real-time color vision enhancement.

Physiological Color Remapping (LMS Color Space)

The base of the Daltonization algorithm relies on modeling human color perception, and this requires looking beyond the device-dependent RGB color space. Color Vision Deficiency is physiologically defined by reduced sensitivity or the absence of one or more cone types (photoreceptors). It utilizes the LMS color space because this color space directly maps to cone responses, so it is the appropriate domain in which to simulate and correct CVD.

A critical step for mathematical accuracy is the initial data processing: digital camera inputs are usually in the non-linear sRGB space. Before any matrix transformation, the input image has to be sRGB-linearized, i.e. gamma correction must be performed in order to ensure that matrix multiplications operate on physically correct, linear light intensity values.

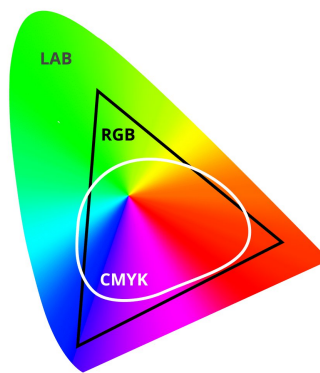
The transformation is defined by:

$$LMS = M_{RGB \rightarrow LMS} \cdot RGB_{Linear}$$

The specific $M_{RGB \rightarrow LMS}$ matrices employed are based on the *Viénot et al. (1999) adaptation of the Hunt-Pointer-Estevez (HPE) cone fundamentals*, providing a scientifically verifiable foundation.

Figure 10

Human Cone Response and the LMS Color Space Model



Note: Adapted from What is RGB, what is CMYK? And how to convert RGB to CMYK? by Team Next

The figure illustrates the relative spectral sensitivities of the Long-wavelength (L), Medium-wavelength (M), and Short-wavelength (S) cone photoreceptors. The significant overlap between the L and M curves highlights the physiological basis for red-green color confusion.

Core Correction Logic and Matrix Operations

The entire correction process is executed in a single pass on the mobile GPU via a *SKSL/GLSL fragment shader*.

CVD Simulation and Error Calculation

The core of the algorithm relies on a calculation of the difference between the original color and the color as seen by the user with the deficiency. The simulation is accomplished by multiplying the original LMS vector by the type-specific Simulation Matrix. Full numerical values for M_{SIM} , M_{CORR} , and the transformation matrices can be found in Appendix K

A) Simulation

$$LMS_{Sim} = M_{Sim} \cdot LMS_{Orig}$$

B) Error Vector

The error vector (**E**), representing the lost spectral information, is calculated

$$E = LMS_{Orig} - LMS_{Sim}$$

C) Compensation and Gain Control

The error vector is shifted into a visible channel (e.g., S-cone axis) via the Correction Matrix (M_{Corr}). The final corrected color (LMS_{Corr}) is a weighted sum of the original color and the compensated error, controlled by the user's Strength Gain Factor ($u_{Strength}$)

$$LMS_{Corr} = LMS_{Orig} + u_{Strength} * (M_{Corr} * E)$$

Figure 13

Daltonization Algorithm Flow Diagram

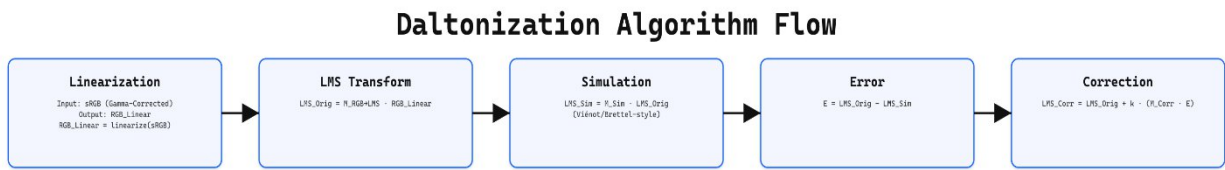


Table 12

Daltonization Algorithm Process Flow

Stage	Description	Purpose
1. Linearization	Converts the input image from non-linear sRGB space (standard for displays) to linear RGB space.	Ensures color mixing and calculations are based on physical light intensity rather than perceptual values, preventing color distortion.
2. LMS Transform	Transforms linear RGB values into the physiological LMS color space.	Maps digital color data to the response curves of human Long, Medium, and Short cone photoreceptors for accurate biological simulation.
3. Simulation	Models the specific Color Vision Deficiency (CVD) by projecting colors onto confusion lines.	Mathematically replicates how the user perceives the image, effectively "flattening" indistinguishable hues (e.g., red/green) into a single tone.
4. Error Calculation	Identifies the specific chromatic information lost due to the deficiency.	Extracts the difference between the original image and the simulated view, isolating the "invisible" spectral data.
5. Correction	Shifts the lost information into a visible spectrum and recombines it with the original image.	Enhances contrast by adding the shifted error back into the image, scaled by the user's intensity setting (\$k\$), to make confused colors distinguishable.

Integration with K-Means

To achieve real-time performance on resource-constrained mobile devices, ReColor performs a hybrid asynchronous pipeline which decouples image analysis from visual rendering. Although the Daltonization algorithm needs per-pixel processing at high frame rates (30+ FPS), K-Means Clustering and CNN-based

classification are computational-intensive tasks that can introduce high latency if executed in a synchronous manner for every frame.

Accordingly, the system architecture assigns these tasks to independent execution threads: while the Analysis Thread periodically samples video frames to refresh color correction parameters given the dominant clusters, the Rendering Thread continuously applies the color correction parameters through an optimized fragment shader. This allows for ensuring that the user interface remains responsive and the video feed remains smooth, even when the system performs such complex scene analysis in the background.

Evaluation Metrics

The evaluation metrics were categorized based on the functional modules of ReColor to ensure that both the screening and enhancement components were appropriately validated. Each module was assessed using domain-specific criteria to guarantee diagnostic accuracy, perceptual reliability, and computational robustness.

Ishihara Module

The Ishihara Module was evaluated using the following metrics to ensure diagnostic validity, consistency with clinical standards, and effectiveness of correction:

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1. *Accuracy (%)*: Measures the proportion of correctly identified responses (normal vs. CVD) compared with the printed Ishihara standard.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP = true positives, TN = true negatives, FP = false positives, and FN = false negatives. Higher accuracy indicates that the system consistently aligns with established Ishihara test outcomes.

2. *Sensitivity and Specificity*: Sensitivity evaluates the ability of the module to correctly identify individuals with CVD, while specificity reflects the accuracy of identifying non-CVD individuals. These are calculated as:

$$Sensitivity = \frac{TP}{TP + FN}, Specificity = \frac{TN}{TN + FP}$$

3. *Cohen's Kappa (κ)* Cohen's Kappa measures the degree of agreement between the system's classifications and expert or clinical diagnoses, adjusting for chance agreement. It is given by:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

Where p_o is the observed agreement between the system and expert ratings, and p_e is the expected agreement by chance. A κ value above

0.80 indicates strong agreement, validating the Ishihara module as a reliable screening tool.

Color Correction Module (Daltonization + K-Means Clustering)

The color correction pipeline was assessed in terms of perceptual accuracy and system responsiveness. To measure perceptual improvement after applying ReColor’s adaptive correction, the CIEDE2000 ΔE formula was used. This metric quantifies color differences between original and corrected outputs as perceived by the human visual system, where lower ΔE values indicate closer alignment to normal color perception and thus higher perceptual accuracy.

$$\Delta E = \sqrt{\left(\frac{\Delta L'}{K_L S_L}\right)^2 + \left(\frac{\Delta C'}{K_C S_C}\right)^2 + \left(\frac{\Delta H'}{K_H S_H}\right)^2 + R_T \left(\frac{\Delta C'}{K_C S_C}\right) \left(\frac{\Delta H'}{K_H S_H}\right)}$$

The components of this formula are interpreted as follows:

- a. **ΔL' (Lightness Difference):** Captures differences in perceived brightness, identifying whether a corrected color appears lighter or darker relative to the original.
- b. **ΔC' (Chroma Difference):** Measures variation in saturation or vividness, reflecting how intense or muted the color appears.

- c. **$\Delta H'$ (Hue Difference):** Represents differences in the actual tone of color (e.g., red, green, blue), accounting for perceptual non-linearities in hue shifts.

- d. **K_L, K_C, K_H (Parametric Factors):** Adjustment constants, typically set to 1 under standard viewing conditions, that control the relative weighting of lightness, chroma, and hue differences.

- e. **S_L, S_C, S_H (Weighting Functions):** Correct for perceptual non-uniformities in the CIELAB color space, normalizing the contributions of lightness, chroma, and hue to reflect human visual sensitivity.

- f. **R_T (Rotation Term):** A correction factor that accounts for complex interactions between chroma and hue differences, particularly in the blue region of the spectrum,

CNN-Based Color Identifier Module

The CNN-based color identifier module was evaluated to ensure classification accuracy, robustness in detecting distinct colors, and its applicability for assisting anomalous trichromats. The following metrics were applied:

1. **Precision** (Positive Predictive Value): Measures the proportion of correctly predicted colors among all predictions made for a given class. High precision ensures that when the system identifies a color, it is reliably correct.

$$Precision = \frac{TP}{TP + FP}$$

where TP = true positives and FP = false positives.

2. **Recall** (Sensitivity): Measures the system's ability to correctly detect all instances of a particular color class. High recall minimizes the risk of misclassifying or missing a target color.

$$Recall = \frac{TP}{TP + FN}$$

where FN = false negatives.

3. **F1 Score** (Harmonic Mean of Precision and Recall): Provides a balanced evaluation by combining precision and recall into a single metric. This is especially important for color classification tasks where class imbalance (e.g., some colors being more common than others) may occur.

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

4. A **confusion matrix** was employed to provide a detailed view of the classification performance across the defined color categories. It

summarizes the counts of true positives, true negatives, false positives, and false negatives for each class, enabling the identification of common misclassifications.

This matrix serves as a complementary evaluation tool to accuracy, precision, recall, and F1-score by offering a class-level perspective of the model’s behavior. The actual results and interpretations of the confusion matrix are presented in Chapter 4.

Sampling Technique

This study utilized purposive sampling, a non-probability sampling technique, to deliberately select participants who could provide the most relevant and meaningful insights into the development and evaluation of ReColor. Purposive sampling was chosen because the study specifically required individuals with Color Vision Deficiency (CVD), a relatively rare condition, as well as experts with specialized knowledge in ophthalmology, optometry, and information technology. Random sampling was not feasible due to the low prevalence of CVD in the general population and the need for expert evaluators with domain-specific expertise.

Through purposive selection, the study ensured that participants represented the primary stakeholders of the system: end-users with CVD, non-CVD individuals for comparative awareness testing, and clinical and technical experts for validation. This method allowed the researchers to gather context-specific data that directly aligned with the study objectives, ensuring that ReColor was both user-centered and technically validated.

Participants of the Study

The study included forty-eight (48) participants, who were intentionally chosen in a way that would give a deep understanding of the clinical, technical, and user-centered points of view regarding the ReColor mobile app. The selection of these participants made it possible for the different points of view to be heard regarding both clinical validation and usability testing, such as from people with Color Vision Deficiency (CVD), regular end-users, vision health practitioners, and information technology (IT) specialists. This purposive blending of participants was in line with the mixed-methods and prospective pilot cohort design of the study, giving equal weight to both qualitative and quantitative validation.

Table 13

Summary of Study Participants of ReColor

Participant Group	No. of Participants	Role in the Study	Primary Purpose	Evaluation Focus
Clinical Participants (CVD)	5	Individuals diagnosed with Color Vision Deficiency (CVD), referred by the Philippine Eye Research Institute	Validate the Ishihara screening and adaptive correction module	Diagnostic accuracy, sensitivity, specificity, Cohen's Kappa
End Users	30	General users (CVD and non-CVD) testing the mobile app	Assess usability, accessibility, and awareness features	Usability scores, accessibility metrics, user satisfaction
Ophthalmologists / Optometrists	3	Licensed eye care professionals providing expert validation	Evaluate clinical soundness and ethical compliance	Expert validation, clinical accuracy, qualitative feedback

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Participant Group	No. of Participants	Role in the Study	Primary Purpose	Evaluation Focus
IT Experts	10	Software and mobile app professionals	Evaluate system performance and technical compliance	Android Core App standards, security, functionality, and compatibility

Clinical Participants (n = 5)

Five participants diagnosed with Color Vision Deficiency were recruited through the Philippine Eye Research Institute (PERI). Their participation was essential for validating the *ReColor* system's screening accuracy and adaptive color correction functionality under authentic visual conditions. This group tested the Ishihara module and color remapping algorithms to determine the perceptual improvement achieved through the app. The number of CVD participants was determined based on PERI's 2025 recommendation, which advised that five participants are sufficient for pilot-level clinical validation, ensuring ethical manageability, individualized testing, and representative feedback within the scope of a controlled study.

General End-Users (n = 30)

Thirty (30) non-CVD participants were included to evaluate the overall usability, accessibility, and performance of the *ReColor* mobile application. These participants represented typical users such as students, professionals, and individuals

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	from the general population who may utilize the app for awareness, educational, and assistive purposes. Their feedback was instrumental in measuring user experience factors such as interface intuitiveness, color differentiation clarity, and satisfaction levels. The inclusion of this group ensured that <i>ReColor</i> adheres to user-centered design principles and complies with the Android Core App Quality Guidelines (Google, 2025). The sample size was selected to provide sufficient diversity and reliability for usability testing and statistical analysis. Ophthalmology Professionals (n = 3) Three (3) ophthalmology professionals, consisting of ophthalmologists and optometrists affiliated with the Philippine Eye Research Institute and collaborating institutions, were consulted to review the diagnostic accuracy of the <i>ReColor</i> screening module. Their expert evaluation provided medical validation of the Ishihara test adaptation and ensured alignment with established color vision screening standards. The inclusion of this group supported the study’s goal of bridging clinical standards with digital accessibility, ensuring that the system functions as both a reliable screening aid and an educational tool. Information Technology Experts (n = 10) Ten (10) IT professionals with expertise in software engineering, artificial intelligence, and human-computer interaction were involved to assess the technical soundness of <i>ReColor</i>. Their evaluation focused on the application’s algorithmic	
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accuracy, system reliability, and adherence to the ISO/IEC 25010 Software Quality Model in terms of functionality, usability, and performance efficiency. Their inclusion provided essential feedback on code quality, data handling, and user interface responsiveness, contributing to the optimization of ReColor’s hybrid pipeline integrating K-Means clustering, Daltonization, and Convolutional Neural Networks (CNNs).

Research Locale

The study was conducted through a distributed research locale design to accommodate the mobile nature of the ReColor system. Since the application operates on Android devices, data collection and usability testing were carried out across multiple locations based on the convenience of the participants. The primary institutional beneficiary and reference point for this study is the Philippine Eye Research Institute (PERI), located in Quezon City, Metro Manila, which serves as the national center for ophthalmic research and vision screening initiatives.

Testing and evaluation were conducted both online and in person, allowing flexibility in participation and ensuring inclusivity regardless of participants’ geographic or logistical constraints. Recruitment of participants was primarily achieved through peer referrals and community recommendations to reach individuals with varying degrees of color vision deficiency.

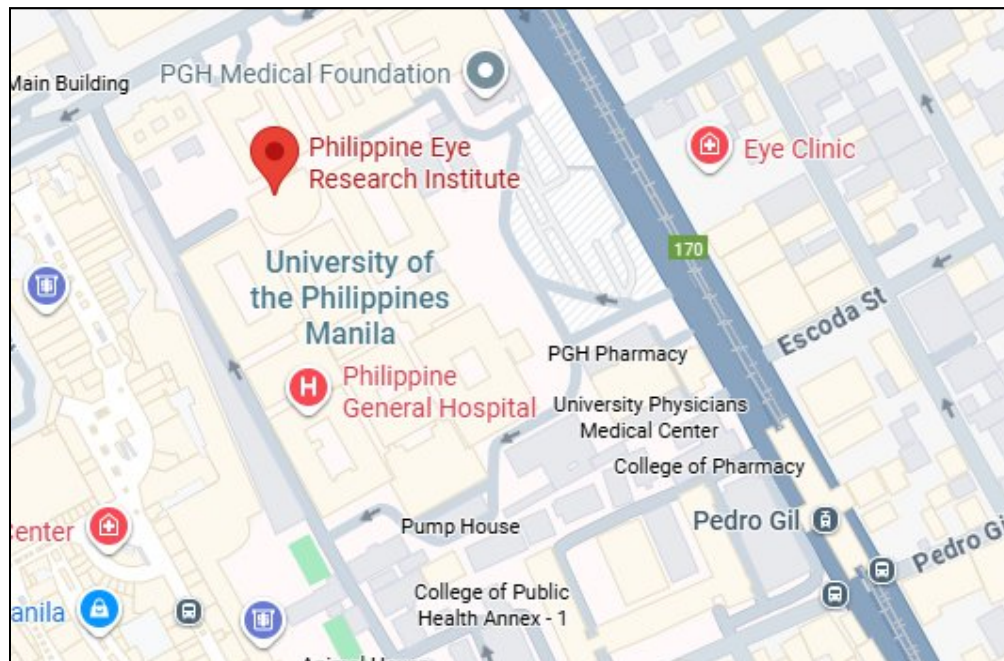
This distributed setup enhanced the ecological validity of the study by capturing real-world variations in lighting, device performance, and environmental conditions that directly influence visual perception and usability outcomes. Figure

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8 presents a map showing the geographical relation of the various participant testing sites to the Philippine Eye Research Institute, highlighting its role as the central institutional point of reference for ReColor's validation.

Figure 12

Geographical location of the Philippine Eye Research Institute (PERI), the primary research beneficiary and evaluation partner for ReColor.



Note: Adapted from Google Maps (2025). Retrieved from <https://www.google.com/maps>

Research Instrument

This study employed specific research instruments for two major phases: problem Identification and system evaluation. Each instrument was designed to capture both user-centered needs and expert validation, ensuring that ReColor was contextually relevant, technically robust, and clinically aligned.

ReColor Mobile Application

The primary research instrument was the *ReColor* mobile application prototype, developed using Python, TensorFlow Lite, and Firebase. It functioned as both a diagnostic and assistive tool for individuals with Color Vision Deficiency (CVD). The prototype incorporated three algorithmic modules, namely K-Means clustering, Convolutional Neural Networks (CNN), and Daltonization, which collectively performed real-time image enhancement and adaptive color remapping. The system allowed users to capture or upload images for color correction, perform Ishihara-based screening tests, adjust adaptive color filters, and receive visual feedback. All generated data were securely stored in Firebase to enable quantitative analysis of performance and user behavior. The application was deployed on Android devices for both in-person and online usability testing.

Digitized Ishihara Test

A digitized version of the Ishihara Color Vision Test was also integrated into the ReColor system to establish baseline and post-intervention measures of visual discrimination ability. The test was administered before and after the use of the system to evaluate its corrective effectiveness. This digital adaptation preserved the original Ishihara plate standards, validated and verified by the Philippine Eye Research Institute, ensuring medical reliability and consistency with recognized diagnostic methods.

Calibration and Screening Limitation Protocol

The system addresses the optical variability of digital displays with a Calibration and Disclaimer Protocol recommended by PERI. In this regard, all applications should automatically enforce a mandatory 'High Brightness' check (minimum 80% luminosity) before screening to approximate the lighting conditions necessary for conventional pseudoisochromatic plates. Also, all test results will be explicitly categorized as 'Preliminary Screening Results' with a permanent disclaimer that the mobile evaluation detects probable color vision deficits without serving as a substitute for the clinical diagnosis of an eye care professional.

Expert Validation Checklist

An expert validation checklist was employed to obtain assessments from ophthalmologists, optometrists, and information technology professionals. The clinical validation component focused on diagnostic accuracy, perceptual enhancement, and compliance with established color vision assessment protocols. The technical validation component was aligned with the Android Core App Quality Standards and evaluated aspects such as performance, usability, responsiveness, accessibility, and system stability. Each criterion was rated using a five-point Likert scale ranging from “Strongly Disagree” to “Strongly Agree.” This provided measurable insights into both clinical soundness and technical functionality.

User Feedback Survey

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A user feedback survey was administered to thirty general users and five individuals with clinically confirmed CVD. The survey instrument focused on usability, accessibility, performance, and awareness impact. It evaluated the intuitiveness of navigation, clarity of design, visibility of color outputs, responsiveness under varying conditions, and changes in awareness regarding color vision deficiency and its professional implications. Prior to implementation, the survey was validated by two academic research advisers and one representative from the Philippine Eye Research Institute. Internal consistency of the responses was assessed using Cronbach's Alpha, with reliability coefficients above 0.80 considered satisfactory.

Interview and Observation Guide

To support the quantitative findings, semi-structured interviews and observational notes were also utilized. The interview guide emphasized participants perceived improvements in color discrimination after using ReColor, the challenges encountered during operation, and recommendations for further system refinement. This qualitative approach provided deeper insight into user perception and reinforced the statistical findings derived from the survey and validation instruments.

Data Gathering Procedure

The data gathering procedure for this study was conducted under the supervision of the Philippine Eye Research Institute (PERI) and in compliance with

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the Lyceum of the Philippines University–Cavite research ethics protocol. It was divided into three phases: preparatory, implementation, and post-evaluation.

During the *preparatory phase*, the researchers coordinated with PERI specialists through an initial consultation held on August 28, 2025, to validate the study’s ethical compliance, technical soundness, and clinical relevance. Research instruments including the ReColor prototype, the digital Ishihara screening tool, and validation checklists were reviewed by both academic and clinical experts. Pilot testing was also conducted with non-participant individuals to refine usability and content before official deployment.

The implementation phase involved three main activities: *system evaluation, user testing, and expert validation*. *System evaluation* verified the operational stability and performance of the ReColor prototype, including its K-Means Clustering, Convolutional Neural Network (CNN), and Daltonization modules. User testing was conducted among thirty (30) participants, including five (5) individuals with clinically confirmed Color Vision Deficiency referred by PERI. Participants performed application-based tasks such as image capture, adaptive color correction, and Ishihara-based screening. Expert validation included three (3) ophthalmologists and ten (10) IT professionals who assessed clinical accuracy and compliance with Android Core App Quality Standards, respectively.

In the *post-evaluation phase*, collected data was compiled, encoded, and analyzed. Quantitative data from Firebase logs and user surveys were subjected to descriptive and inferential analysis to measure system effectiveness and usability, while qualitative feedback from interviews was analyzed thematically to capture user

experience insights. All participant information was anonymized using coded identifiers, and data were securely stored in accordance with the Data Privacy Act of 2012 (RA 10173).

System Development Process

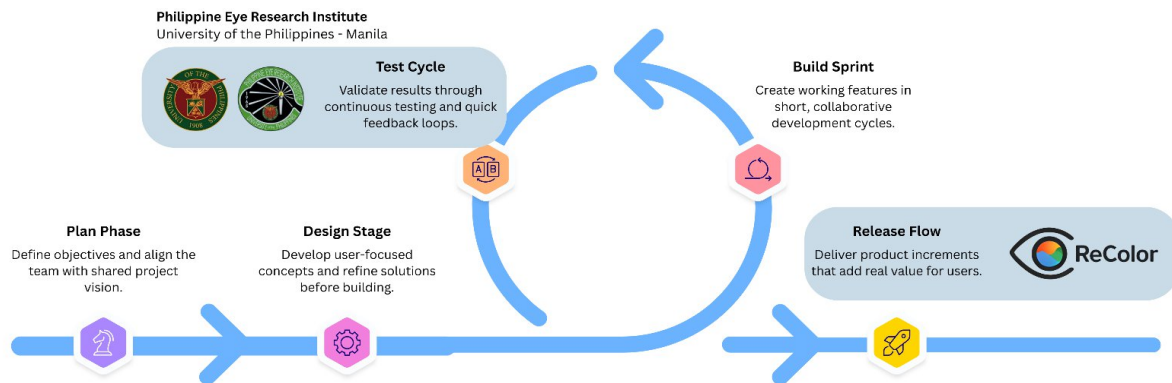
This study employed a Modified Agile–Spiral methodology to accommodate both the dynamic requirements of system development and the structured demands of academic research. While the standard Agile framework emphasizes daily stand-ups and rapid software deliverables, the modified approach extended its scope by incorporating weekly sprint meetings documented through official minutes of the meeting (MoM) led by the scrum lead. Each sprint cycle produced not only functional software increments but also research deliverables, such as conceptual frameworks, survey instruments, and academic documentation.

Figure 13

Modified Agile–Spiral Diagram of ReColor



Modified Agile–Spiral Diagram

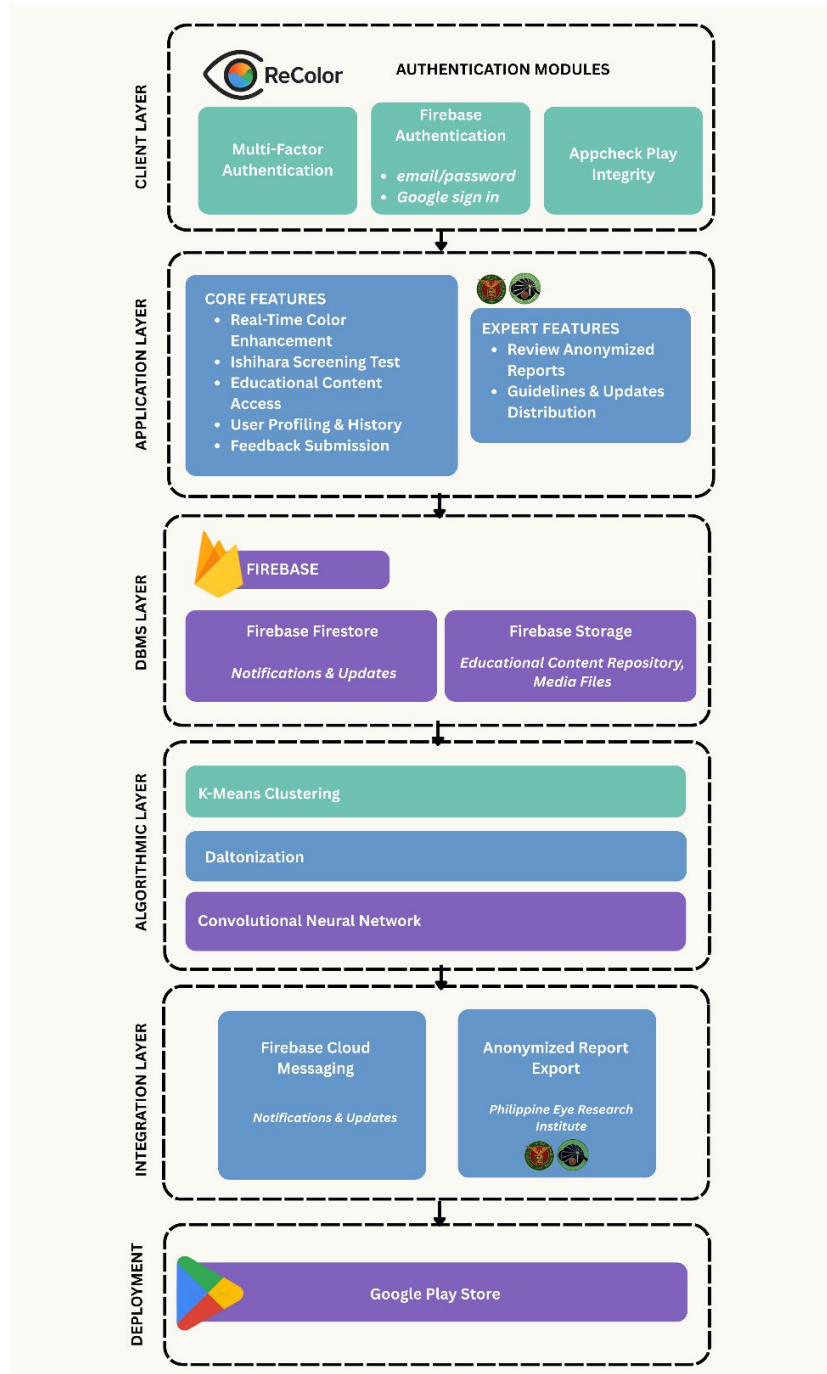


As reflected in the Agile Workflow Diagram (Figure 6), the Spiral model's emphasis on iterative risk assessment was adapted through structured feedback loops. Bi-monthly consultations with the Philippine Eye Research Institute (PERI) were integrated into the Test Cycle, ensuring clinical accuracy and alignment with vision science standards. At the same time, IT specialists provided ongoing technical evaluations during sprint reviews and retrospectives, reinforcing the technical robustness of the system. By embedding these validation checkpoints into the iterative cycles of planning, designing, building, and releasing, the methodology ensured that each development cycle addressed both technological feasibility and research rigor. This hybrid process ultimately aligned with the project's dual objectives of practical system implementation and scholarly contribution.

System Architecture

Figure 14

System Architecture of ReColor



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	<i>Client Layer.</i> This layer functions as an authentication module, ensuring secure access to ReColor’s features. Users can create accounts using email and password credentials or through Google Sign-In, both managed by Firebase Authentication. To enhance security, Multi-Factor Authentication (MFA) is enabled, while AppCheck Play Integrity provides an additional safeguard by verifying that the application is running on genuine devices with authorized signing keys from the Google Play Console. This setup ensures that authentication processes remain secure and validated before granting users access to the system’s features and backend services, including Firebase Firestore, Storage, and Cloud Messaging. <i>Application Layer.</i> Once authenticated, users are directed to role-based interfaces according to their classification as either CVD users, non-CVD users, or institutional experts such as the Philippine Eye Research Institute (PERI). CVD and non-CVD users access the app’s core features, including real-time color enhancement, Ishihara screening tests, educational content, user profiling, and feedback submission. The system implements a secure, web-based Institutional Admin Dashboard as a centralized data repository to support the research mandates of PERI. This interface aggregates anonymized user inputs collected from the Quick Survey and Ishihara Screening modules of the mobile application. <i>Database Management System (DBMS) Layer.</i> Firebase Firestore serves as the primary data repository, storing structured information such as user profiles, test histories, and screening outcomes. Timestamped entries allow longitudinal tracking of user progress, while real-time synchronization ensures immediate feedback and updates. Firebase Storage is employed for educational content repositories and	
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multimedia files, providing scalable and secure storage for non-text data. Together, these services support both personalized user experiences and broader institutional data needs, maintaining consistency, reliability, and scalability.

Algorithmic Layer. At the heart of ReColor’s novelty lies its hybrid adaptive color correction pipeline. K-Means clustering is employed to efficiently group pixels and reduce computational overhead, Daltonization remaps colors along confusion lines to enhance perceptual clarity, and Convolutional Neural Networks (CNNs) perform adaptive feature extraction to deliver personalized, real-time color transformations tailored to individual CVD profiles. These components work in tandem to provide responsive visual enhancements that go beyond traditional filtering techniques.

Integration Layer. This layer ensures communication between the application and external entities. Firebase Cloud Messaging (FCM) enables notifications and updates for users, including screening reminders, awareness prompts, and career guidance information. In parallel, anonymized reports and aggregated screening data are exported to PERI, ensuring alignment with clinical research and national vision screening policies. This bi-directional integration strengthens the app’s role not only as a personal assistive tool but also as a research-informed, public health-aligned solution.

Deployment Layer. Finally, the ReColor application is deployed through the Google Play Store, ensuring accessibility, scalability, and compliance with Android platform requirements. Play Store distribution provides streamlined updates, device

compatibility checks, and widespread user reach, making ReColor both technically robust and practically implementable in the Philippine context.

Hardware Specifications

To ensure ReColor runs smoothly and delivers a responsive, real-time vision assistive experience, the following hardware specifications are recommended. ReColor is implemented in React Native with native camera access (react-native-vision-camera), OpenCV-based image processing, and TensorFlow Lite for on-device inference. While the application is designed to function on a wide range of mobile devices, meeting the recommended specifications will help achieve optimal performance and user experience.

a. Processor (CPU)

Minimum: Midrange ARM-based SoC with at least four cores (example: Qualcomm Snapdragon 400 series or equivalent).

Recommended: Midrange to upper-midrange ARM SoC with 6+ cores for better parallel processing and image pipeline throughput (example: Snapdragon 600/700 series or equivalent). The CPU should support efficient background threads for clustering and UI tasks.

b. Neural/Graphics Acceleration (NPU/GPU)

Minimum: Device should support Android NNAPI or equivalent for hardware-accelerated inference. If no NPU is available, a GPU delegate (OpenGL ES 3.0+) is required for acceptable TFLite performance.

Recommended: Device with an integrated NPU or modern GPU (e.g., Mali G-series, Adreno 600 series) to run TensorFlow Lite models with hardware acceleration. This reduces latency and power consumption for CNN inference.

c. Memory (RAM)

Minimum: 4 GB RAM.

Recommended: 4 GB to 6 GB RAM to permit concurrent camera frame buffering, clustering, model inference, and UI rendering without swapping or stutters.

d. Storage

Minimum: 1 GB of free storage dedicated to the app, cached frames, and user test results.

Recommended: Devices with 32 GB total internal storage or higher and at least 1.5 GB free to accommodate models, cached images, user profiles, and future updates. Typical TFLite model sizes may range from tens to a few hundred megabytes depending on quantization and complexity.

e. Camera and Imaging Sensor

Minimum: Single rear or front camera capable of at least 720p (1280 × 720) capture at 30 frames per second, with autofocus and exposure control. The camera API must support low-latency access (Camera2 or VisionCamera integration).

Recommended: 1080p-capable camera with good low-light performance, hardware autofocus, and adjustable exposure. Support for high frame rates (30–60 fps) improves responsiveness for live correction. Prefer cameras that allow manual control of exposure and white balance where possible.

f. Display and Interaction

Minimum: 720p display resolution for clear UI rendering and legible educational content. Haptic motor and audio output are required for multimodal feedback.

Recommended: Full HD (1080p) display for clearer visualization of corrected colors, instructional content, and career-awareness materials. Support for touch gestures and accessible UI (large targets, scalable fonts) is expected.

g. Battery and Thermal Considerations

Recommended: Device battery capacity of 3000 mAh or greater to allow extended use during testing. Real-time video processing and inference are power intensive, so thermal throttling can affect performance. Devices with efficient thermal design are preferred.

h. Sensors and Peripherals

Recommended: Ambient light sensor to support adaptive correction under varied lighting, microphone for optional voice guidance, and vibration motor for haptic cues. GPS is not required. Bluetooth is optional for future accessory support.

i. Connectivity and Offline Behavior

Minimum: Wi-Fi or cellular (4G/LTE) for synchronization of anonymized test results and updates. Core features, including screening and real-time correction, must be designed to operate offline.

Recommended: Stable internet connection for initial onboarding, optional cloud backup, and data sharing with PERI through Firebase. Support for intermittent connectivity and background sync is advised.

j. Operating System and Compatibility

Minimum: Android 8.0 (API level 26) to ensure broad device coverage and NNAPI availability. The app will be tested for compatibility on Android 8.0 to Android 15 (API level 35).

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	Recommended: Android 11+ for best hardware acceleration support and modern API availability. iOS support may be provided where equivalent hardware acceleration (Core ML, Metal) is available. k. Development and Test Device Matrix (for benchmarking) At least three representative devices should be used during development and evaluation: a. Low-end device (minimum spec) to measure worst-case performance. b. Midrange device (recommended target) for primary optimization and user testing. c. High-end device to measure best-case performance and upper bounds.	
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Data Analysis

Data analysis for this study was carried out through a combination of quantitative, qualitative, and cohort-based evaluation methods to comprehensively assess the effectiveness, usability, and clinical alignment of the *ReColor* mobile application.

For the *quantitative analysis*, system performance data collected through Firebase logs and algorithm outputs were processed using statistical software to measure the accuracy and efficiency of the K-Means Clustering, Convolutional Neural Network (CNN), and Daltonization modules. Metrics such as ΔE (CIEDE2000) for perceptual accuracy, Structural Similarity Index (SSIM) for visual fidelity, and confusion matrix-based accuracy were computed. Usability and accessibility ratings obtained from the thirty (30) end-user participants were analyzed using descriptive statistics, while inferential tests such as paired *t*-tests and correlation analysis were used to compare pre- and post-correction scores from the Ishihara-based screening.

For the *qualitative analysis*, feedback from semi-structured interviews and focus group discussions with users, ophthalmologists, and IT experts was transcribed and analyzed through thematic coding. Emerging themes focused on user experience, accessibility, clinical credibility, and visual comfort. These qualitative insights supported the interpretation of quantitative results by highlighting usability challenges and areas of perceptual improvement from the participants' perspective.

The *pilot cohort analysis* was conducted on the five (5) clinically diagnosed Color Vision Deficiency participants referred by the Philippine Eye Research Institute. Their responses to Ishihara tests before and after using ReColor’s adaptive correction feature were compared longitudinally to assess perceptual enhancement and screening consistency.

All data underwent validation and cross-verification between quantitative results and expert evaluations. Statistical significance was set at $p < 0.05$ for inferential analyses, ensuring objective interpretation. Data integrity and confidentiality were maintained throughout the process in compliance with the Data Privacy Act of 2012 (RA 10173).

Ethical Consideration

In line with recommendations from the Philippine Eye Research Institute (PERI, 2025), we employ strict ethical considerations throughout the study. The following measures were implemented to protect participants, preserve data integrity, and ensure responsible scientific conduct during the development, deployment, and evaluation of ReColor.

1. *Data privacy and confidentiality.* Participant privacy and confidentiality are principal concerns given the collection of sensitive vision and health-related information. All personally identifiable information will be removed or replaced with a unique participant code prior to analysis and reporting. Data in transit will be protected using TLS 1.2 or higher and data at rest will be encrypted using AES-256.

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	Firestore security rules and App Check will be configured to restrict access to authenticated applications only. Aggregate and anonymized datasets may be shared with collaborating institutions such as the Philippine Eye Research Institute for approved research purposes only. Individual-level data will not be disclosed without explicit participant consent. 2. <i>Participant Code System and Pseudonymization.</i> To preserve confidentiality and uphold ethical research standards, every participant will be assigned a unique participant code upon enrollment. These codes will replace personal identifiers in all data records, logs, transcripts, images, spreadsheets, and published materials. The following procedures will be implemented to ensure robust privacy protection: a. <i>Code Format.</i> Participant codes will follow a consistent structure, for example: REC-2025-001, where REC denotes the project shorthand, 2025 indicates the year of enrollment, and 001 represents a zero-padded sequence number. b. <i>Linkage Key.</i> A separate mapping file linking participant names to their corresponding codes will be stored independently from all research data in an encrypted file. Access to this linkage key will be restricted to authorized personnel only and managed by the project data steward. c. <i>Secure Storage.</i> The linkage key will be encrypted using industry-standard encryption algorithms and stored on a secure server or encrypted drive. Access controls will utilize role-based permissions and multi-factor authentication for additional protection.	
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d. *Access Logging and Audit.* All instances of access to the linkage key will be automatically logged. An access audit trail will be maintained to record user identity, access time, and purpose of retrieval.

e. *Use in Outputs.* All analyses, transcripts, figures, tables, and appendices will exclusively reference participant codes. No direct identifiers (e.g., names, contact information) will appear in any thesis outputs or related publications.

f. *Retention and Deletion.* The linkage key will be retained only for the duration necessary to complete the study objectives and any clinical referral obligations. Upon verified participant withdrawal or at the end of the retention period, all identifiable linkage files will be permanently deleted within 30 days. Fully anonymized datasets may be retained for up to five years for reproducibility and secondary analysis.

g. *Images and Media.* Any images or visual data used for algorithm validation will be renamed using participant codes, stripped of metadata (e.g., EXIF data), and stored securely. Faces or identifiable features will be blurred or removed unless explicit written consent for identifiable media has been obtained.

3. *Compliance with law and institutional policy.* The research will comply with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173), university IRB or ethics committee requirements, and other applicable institutional policies. Where data are shared across borders, the research team will follow relevant legal and contractual data transfer safeguards.

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4. *Informed consent and voluntary participation.* Participants will receive clear written information about the study purpose, procedures, expected duration, potential benefits, and risks. Consent forms will explain how data are stored, who will have access, how participant codes are used, and how participants can withdraw.

Participation is voluntary and participants may withdraw at any time without penalty.

The consent form will be available in English and Tagalog and will include the following short statement about pseudonymization: "To protect your privacy we will assign you a unique participant code. Your name will be stored separately from your survey responses and test results. The linkage file that maps names to codes will be encrypted and accessible only to authorized researchers. You may request deletion of your identifiable data at any time."

5. *Screening and clinical referral.* The Ishihara module is a screening tool and not a clinical diagnostic instrument. Participants who screen positive will receive guidance and referral information to seek confirmatory clinical evaluation from qualified eye care professionals, including contacts at the Philippine Eye Research Institute when appropriate. The research team will not provide medical diagnoses.

6. *Minimization of risk.* The study involves minimal physical risk. Potential psychosocial risks include distress following a positive screening result. To mitigate this, the consent process will outline possible outcomes and provide referral resources. The research team will document and respond to any adverse events.

7. *Image and media handling.* Camera captures or uploaded images used in algorithm validation may include identifiable content. Only images explicitly consented for research use will be stored. Identifiers will be removed or blurred

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	unless separate consent is obtained. All image data will be subject to the same encryption and access control measures as other sensitive data. 8. <i>Cybersecurity and data integrity.</i> Technical safeguards to protect the system and participant data will include secure authentication via Firebase Authentication, App Check, least-privilege access roles, encrypted storage, routine software updates, and scheduled security audits. Critical operations and access events will be logged for accountability and forensic review if needed. A data breach response plan is in place to notify the IRB and affected participants within 72 hours of detection. 9. <i>Bias, fairness, and model transparency.</i> The research team will mitigate algorithmic bias by using diverse training and validation datasets and by reporting model performance across demographic subgroups where feasible. Model conversion, quantization, and compression steps performed for TensorFlow Lite deployment will be documented. Model limitations and known failure modes will be disclosed to participants and expert evaluators. 10. <i>Intellectual property and licensing.</i> The study will use third-party libraries, datasets, and test materials in accordance with their licenses. Copyrighted test materials such as Ishihara plates will be used only with appropriate permissions or under fair-use provisions and will be properly credited. 11. <i>Social impact, inclusion, and non-discrimination.</i> The project will design inclusive interfaces and educational content and will avoid discriminatory recruitment or evaluation practices. Recruitment and communication will accommodate accessibility needs. Career guidance content will be factual and non-stigmatizing. COLLEGE OF INFORMATION TECHNOLOGY AND COMPUTER SCIENCE	
		PAGE *

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	12. <i>Dual-use and misuse prevention.</i> To reduce dual-use risks, access to raw datasets will be restricted, and any external sharing will require data-use agreements that prohibit misuse. Sensitive operational details that could facilitate misuse will be handled with care while maintaining scientific transparency. 13. <i>Accountability and integrity.</i> The research team commits to honest and accurate reporting of methods and results. Any conflicts of interest will be disclosed. A record of protocol changes, ethical approvals, consent forms, and minutes of stakeholder consultations including PERI meetings will be maintained and made available to the IRB upon request. 14. <i>Collaboration ethics and credit.</i> All collaborators, advisors, and participants will receive appropriate acknowledgment. Authorship will follow accepted academic conventions. Any financial or professional relationships that could influence the study will be declared. 15. <i>Data retention and participant rights.</i> The consent form will state the data retention schedule and provide instructions for participant requests such as data access, correction, or deletion. The procedure for honoring deletion requests will be described and executed within the specified timeframes. 16. <i>Data steward and point of contact.</i> A designated project data steward will manage encryption keys, linkage files, and data access requests. Contact details for privacy and data deletion inquiries will be included in the consent form and the study documentation.	
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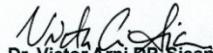
Appendix A
ReColor Communication Letters

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Lyceum of the Philippines University – Cavite
College of Information Technology and Computer Science

ReColor Letter of Intent


Dr. Victor Arni DP. Sicam *Received*
Director
Philippine Research Institute

5F SOJR Bldg.,
UP-PGH Compound,
Ermita, Manila 1000

Dear Dr. Sicam,

Greetings in Veritas et Fortitudo Pro Deo et Patria!

We, **Mark Risen B. Caparas, Charles Justine B. Rull, Johann Nikkolai L. Cepeda, and Stephen L. Lacsá** from the Lyceum of the Philippines University – Cavite, are currently undertaking a thesis project entitled **"ReColor: A Mobile Vision Assistive Tool for Color Vision Deficiency Using Adaptive Daltonization and Clustering Algorithms."**

The purpose of our study is to design, develop, and evaluate "ReColor," a mobile assistive technology aimed at helping individuals with color vision deficiency by utilizing innovative adaptive daltonization and clustering algorithms. We are honored that our thesis has been selected as one of the beneficiaries under your esteemed projects.

We would like to express our sincere gratitude to your humble office and as well to Ms. Em-Em for the enlightening meeting we had with her last August 28, 2025, which provided us valuable insights and encouragement to pursue this partnership for our thesis.

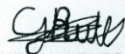
In line with this, we respectfully request an appointment with you and your team on **September 4, 2025, at 11:00 AM, or at your most convenient time**, to formally document the partnership and to seek your expert input that would greatly enrich the study. We believe that your guidance and collaboration will be instrumental in achieving the objectives of this research. Rest assured that all information shared will be treated with the utmost confidentiality and used strictly for academic purposes.

Thank you very much for your time and consideration. We look forward to your favorable response.

Sincerely yours,



MR. MARK RISEN B. CAPARAS
Team lead



MR. CHARLES JUSTINE B. RULL
Back-end Developer

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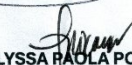


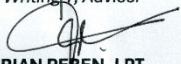

MR. JOHANN NIKKOLAI L. CEPEDA
Front-end Developer


MR. STEPHEN L. LACSA
User Interface/ Experience Designer

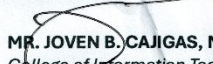
Noted by:


MR. ARCELL R. HADLOCON, MIT, LPT
Thesis Advisor


MS. ALYSSA PAULA POCAAN, MIS
Thesis Writing 1, Adviser


DR. JERIAN PEREN, LPT
Program Chair, Computer Science

Approved by:


MR. JOVEN B. CAJIGAS, MIT
College of Information Technology and Computer Science,
Dean

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE



Mark Risen Caparas

To: upm-peri@up.edu.ph

Cc: Johann Nikkolai Cepeda; Charles Justine Rull; Stephen Lacsa; eblamento@up.edu.ph; Arcell Hadlocon



Reply

Reply all

Forward



Wed 9/3/2025 3:15 PM



2 attachments (538 KB)  Save all to OneDrive - Lyceum of the Philippines University  Save All Attachments

Dr. Victor Arni DP. Sicam
Director
Philippine Research Institute

5F SOJR Bldg.,
UP-PGH Compound,
Ermita, Manila 1000

Dear Dr. Sicam,

Greetings in Veritas et Fortitudo Pro Deo et Patria!

We, **Mark Risen B. Caparas, Charles Justine B. Rull, Johann Nikkolai L. Cepeda, and Stephen L. Lacsa** from the Lyceum of the Philippines University – Cavite, are currently undertaking a thesis project entitled “**ReColor: A Mobile Vision Assistive Tool for Color Vision Deficiency Using Adaptive Daltonization and Clustering Algorithms.**”

The purpose of our study is to design, develop, and evaluate “ReColor,” a mobile assistive technology aimed at helping individuals with color vision deficiency by utilizing innovative adaptive daltonization and clustering algorithms. We are honored that our thesis has been selected as one of the beneficiaries under your esteemed projects.

We would like to express our sincere gratitude to your humble office and as well to Ms. Em-Em for the enlightening meeting we had with her last August 28, 2025, which provided us valuable insights and encouragement to pursue this partnership for our thesis.

In line with this, we respectfully request an appointment with you and your team on **September 4, 2025, at 11:00 AM, or at your most convenient time**, to formally document the partnership and to seek your expert input that would greatly enrich the study. We believe that your guidance and collaboration will be instrumental in achieving the objectives of this research. Rest assured that all information shared will be treated with the utmost confidentiality and used strictly for academic purposes.

Thank you very much for your time and consideration. We look forward to your favorable response.

Sincerely yours,

- MR. MARK RISEN B. CAPARAS**
Team lead
- MR. CHARLES JUSTINE B. RULL**
Back-end Developer
- MR. JOHANN NIKKOLAI L. CEPEDA**
Front-end Developer
- MR. STEPHEN L. LACSA**
User Interface/ Experience Designer

Appendix B
ReColor Minutes of the Meeting

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE



Lyceum of the Philippines University – Cavite College of Information Technology and Computer Science

ReColor Minutes of the Meeting

August 30, 2025, 9:00 – 10:00 pm

Agenda

- Sprint Meeting 1 (August 30, 2025)

Attendees:

1. Mark Risen Caparas – Team Lead
2. Stephen Lacsá – UI/UX Designer
3. Charles Justine Rull – Backend Developer
4. Johann Nikkolai Cepeda – Front-end Developer
5. Justine Deluana – Honorary Backend Developer

Summary:

The thesis team convened to discuss the development of an application aimed at addressing color perception issues, particularly for colorblind users. They agreed to communicate in English for improved AI transcription and emphasized the importance of selecting an inclusive color palette. A suggestion was made to digitize the Ishihara test, while Stephen proposed incorporating a VR function to assist users in identifying colors.

Tasks were assigned to front-end developers Johann and Stephen, with a focus on researching the app's capabilities. The team explored various forms of color vision deficiency, such as protanopia and deuteranopia, and discussed the limitations of current interventions and diagnostic tools, including the Ishihara test, highlighting the need for tailored approaches and further resources to support individuals with color vision deficiencies.

Summary:

The thesis team convened to discuss the development of an application aimed at addressing color perception issues, particularly for colorblind users. They agreed to communicate in English for improved AI transcription and emphasized the importance of selecting an inclusive color palette. A suggestion was made to digitize the Ishihara test, while Stephen proposed incorporating a VR function to assist users in identifying colors.

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Chapters & Topics:

Thesis Team Discussion

In the first thesis team meeting, it was highlighted by a conference room participant that there is a need for English communication to facilitate AI transcription. The discussion included color palette choices for front-



end developers, focusing on accessibility for colorblind users, and the potential digitization of the Ishihara test. Stephen suggested incorporating a VR function that scans surroundings to assist users in identifying colors.

- * Strategies for accommodating colorblind users in app design.
- * Discussion on the color palettes for the app's front-end development.
- * Exploration of VR technology for color identification.

Discussion on Color Perception and Adaptation Techniques

Participants discussed the challenges faced by individuals with color perception deficiencies and the adaptations they employ. They mentioned devices like colorblind spectacles that enhance contrast but do not cure the condition. The conversation also touched on the impact of certain medications on color perception and the potential for therapies, including gene therapy.

Color Vision Deficiency and Interventions

The conversation addressed the complexities of color vision deficiency, detailing types like protanopia, which affects red perception, and deuteranopia, which impacts green perception. Participants discussed the limitations of current interventions and emphasized the importance of understanding individual needs for effective solutions. They noted that coping strategies are often employed by those with color vision deficiencies, as established therapies are lacking.

Diagnostic Tools for CBD Severity Detection

The group examined the limitations of the Ishihara test as a gold standard for diagnosing color vision deficiency (CVD). It was pointed out by a conference room participant that while the test is widely used, it has issues related to standardization and the need for proper calibration of devices. Participants discussed how different settings, such as brightness, can impact the reliability of test results.

- * Limitations of current diagnostic tools for color vision deficiency.

Action Items:

- * The conference room participant will share the screen to present the thesis slides during the meeting.
- * Stephen Lacsá will explore the possibility of implementing a VR function that scans surrounding areas for color identification.
- * The conference room participant will request Johann and Stephen to work on the front-end development of the app.
- * The conference room participant will conduct research to determine if the Ishihara test can be digitized for the app.

Key Questions:

- * What are the specific color palettes that should be used for the front-end development?
- * What limitations exist in the current diagnostic tools for color vision deficiency?

Assignments:

- Sept 3 - Chap 1&2 initial draft submission
- Sept 4 - Follow-up meet with client @Sta. Cruz
- Mark - Finalize Initial Draft
- Nikkolai & Prince - Work on Initial look of App.
- Charles & Justine - Conceptualize the Initial backbone of the app.

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Noted by,

MARK RISEN CAPARAS

Agile/Scrum Lead

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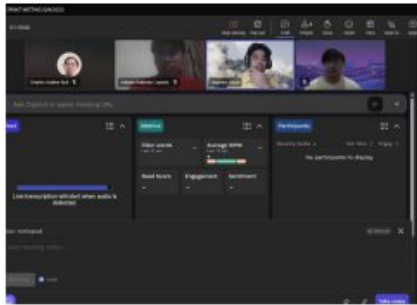
College of Information Technology and Computer Science



Assignments:

- o Mark – Initial Draft of Chapter 3 and Research for applicable methods for ReColor
- o CJ – Refine the K means Algo until the end of the month and Collaborate with Front-end
- o Stephen – Align the Panels to the Use Case Diagram
- o Nikkolai – Work on React Native environment and Implementation mockups

Documentation:



Noted by,

MARK RISEN CAPARAS

Agile/Scrum Lead

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Lyceum of the Philippines University – Cavite College of Information Technology and Computer Science

ReColor Minutes of the Meeting

Agenda

- Sprint Meeting 3 (September 14, 2025)

Attendees

1. Mark Risen Caparas – Team Lead
2. Stephen Lacsa – UI/UX Designer
3. Johann Nikkolai Cepeda – Front-end Developer
4. Charles Justine Rull – Backend Developer
5. Justine Deluana – Honorary Backend Developer

Summary

The thesis team met to discuss the progress of the **ReColor application** and the accompanying paper. Updates covered development on the UI/UX, front-end, and backend components, as well as the ongoing writing of Chapter 3 (Methodology).

- **Paper Progress:** Chapters 1 and 2 finalized and passed AA detection. Chapter 3 is underway, focusing on methodology and technical details.
- **UI/UX (Stephen):** Completed the test tab flow (Pillar 1). Currently working on the Learn tab flow (Pillar 2). Upcoming work will cover profile management, history, theme selector, data insights, and feedback form (Pillar 3).
- **Front-end (Johann):** Completed a semi-prototype of the old Figma design and created a GitHub repository. Currently testing React Native with Expo and implementing new pillar designs. Aim to complete remaining screens by next sprint.
- **Backend (Charles):** Progress limited due to internship workload, but development is expected to resume after internship completion.
- **Backend (Justine):** Focused on image processing using a CNN. Implemented feature suggestion to capture details for color differentiation. Currently debugging and refining the training process.
- **Internship Updates:** Johann completed internship last month. Charles and Justine expected to finish by Thursday.



Chapters & Topics
Paper Progress and Chapter Updates

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- **Mark** – Oversee Chapter 3 drafting in sync with dev updates.

- Key Questions**
- How can Chapter 3's technical details be aligned with ongoing development?
 - What tools will ensure front-end integration with UI/UX mockups?
 - How can CNN performance be improved for accurate color differentiation?

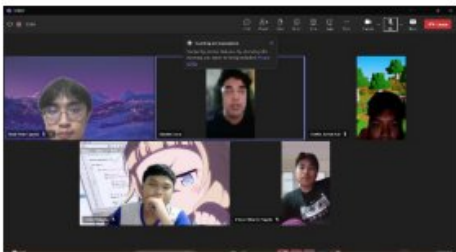
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- Assignments (to be completed within Sprint 3)**
- Johann – Finalize repository sharing.
 - Stephen – Present progress on Learn tab development.
 - Charles – Provide backend update post-internship.
 - Justine – Report CNN debugging results and enhancements.
 - Mark – Submit draft of Chapter 3 methodology section.

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Noted by,

MARK RISEN CAPARAS
Agile/Scrum Lead

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Lyceum of the Philippines University – Cavite College of Information Technology and Computer Science

ReColor Minutes of the Meeting

Agenda

- Sprint Meeting 4 (September 22, 2025)

Attendees

1. Mark Risen Caparas – Team Lead
2. Stephen Lacsa – UI/UX Designer
3. Justine Deluana – Honorary Backend Developer

Summary

The team met to discuss updates on the paper, UI/UX, and backend progress. The focus was on transitioning from conceptualization to actual implementation of the ReColor system. Mark highlighted the need for significant coding progress due to approaching deadlines, setting a target of 20% implementation for the front-end and 10% for the backend.

- **Paper Progress (Mark):** Chapters 1 and 2 finalized. Chapter 3 drafting underway but dependent on front-end and back-end implementation. Diagrams prepared. Plans to polish Chapter 3 once development progresses.

- **UI/UX (Stephen):** Completed Pillar 3 (Learn Tab flow, Educational Hub, Article Reader, user profile, history screens). Completed Pillar 4 (Profile Dashboard, Test History, Theme Selector, Anonymized Data Insights, User Feedback). Major revisions finished. Next steps: minor cleanups and coding collaboration with Johann.

- **Backend (Justine, for Charles):** CNN for image processing implemented at ~50% success with OpenCV, Python, and TensorFlow. Device limitations prevented full training. Plans to integrate Daltonization algorithm first, followed by K-Means. Backend integration with React Native planned.

- **Front-End (Johann):** Absent, no updates reported.

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Chapters & Topics

Paper Progress and Chapter Updates

- Chapters 1 & 2 finalized and ready.
- Chapter 3 (Methodology) in progress; dependent on implementation updates.
- Paper refinement ongoing with diagrams completed.

UI/UX Development (Stephen)

- Pillar 3 (Learn Tab Flow) – Completed: Education Hub, Article Reader, profile, history.
- Pillar 4 (Profile & Management) – Completed: dashboard, test history, theme selector, anonymized insights, feedback.
- Next step – minor cleanups and coordination with Johann for front-end coding.

Front-End Development (Johann)

- Absent during meeting.
- Pending: translate Figma designs into React Native and begin active coding implementation.

Backend Development (Charles & Justine)

Documentation:



Noted by,

MARK RISEN CAPARAS

Agile/Scrum Lead

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Lyceum of the Philippines University – Cavite
College of Information Technology and Computer Science

ReColor Minutes of the Meeting

September 29, 2025, 2025, 9:00 – 10:00 pm

Agenda

- Sprint Meeting 5

Attendees

1. Mark Risen Caparas – Team Lead
2. Stephen Lacsa – UI/UX Designer
3. Justine Deluana – Honorary Backend Developer
4. Charles Justine Rull – Backend Developer
5. Johann Nikkolai Cepeda – Front-End Developer

Summary

The team met to discuss progress updates on the paper, UI/UX, backend, and frontend development. The focus was on aligning the research paper with system implementation while addressing technical challenges in backend model training and frontend integration. The team also finalized next steps for the upcoming sprint and external requirements.

- Paper Progress (Mark): Chapters 1 and 2 finalized. Chapter 3 drafting ongoing, dependent on implementation updates. Emphasis on syncing research and technical milestones.
- UI/UX (Stephen): Completed all major revisions. Transitioning toward minor cleanup and collaboration with Johann for React Native integration.
- Backend (Charles & Justine): CNN image processing tested with ~50% success using OpenCV, Python, and TensorFlow. Due to device limitations, plan to focus first on integrating Daltonization algorithm, followed by K-Means clustering. Backend will be prepared for eventual React Native integration.
- Front-End (Johann): Will begin translation of finalized Figma designs into React Native. Pending initial coding progress.

Chapters & Topics

Paper Progress and Chapter Updates

- Chapters 1 & 2 – Finalized and ready.
- Chapter 3 (Methodology) – In progress; dependent on development updates.
- Paper refinement ongoing; diagrams prepared.

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UI/UX Development (Stephen)

- Completed Pillar 3 (Learn Tab Flow: Education Hub, Article Reader, profile, history).
- Completed Pillar 4 (Profile & Management: dashboard, test history, theme selector, anonymized insights, feedback).
- Next step – minor cleanups and collaboration with Johann for React Native front-end coding.

Front-End Development (Johann)

- Pending – translation of Figma designs into React Native.
- Next step – begin active coding implementation to meet target milestones.

Backend Development (Charles & Justine)

- CNN image processing tested at ~50% accuracy using OpenCV, Python, and TensorFlow.
- Device limitations prevented full model training.
- Plan to first integrate Daltonization algorithm, then K-Means clustering.
- Integration with React Native planned in upcoming sprints.

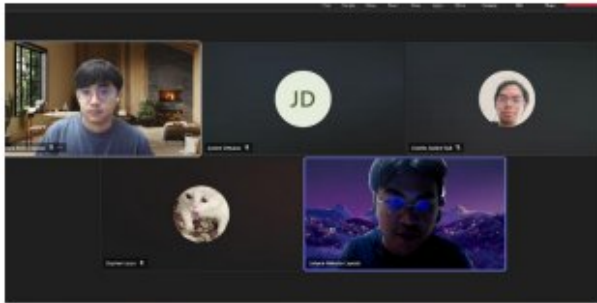
Assignments (to be completed within Sprint 5)

- Paper – Continue refinement of Chapter 3 once development progresses (Mark).
- UI/UX – Minor revisions and React Native collaboration (Stephen & Johann).
- Front-End – Begin implementation of designs in React Native (Johann).
- Backend – Integrate Daltonization algorithm and prepare for K-Means integration (Charles & Justine).
- External – Attend UP Manila on October 9 (Team).

Documentation:



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Noted by,

MARK RISEN CAPARAS

Agile/Scrum Lead

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Lyceum of the Philippines University – Cavite College of Information Technology and Computer Science

ReColor Minutes of the Meeting

Agenda

- Sprint Meeting 6 (October 7, 2025)

Attendees

1. Stephen Lacsa – UI/UX Designer
2. Johann Nikkolai Cepeda – Front-End Developer
3. Charles Justine Rull – Backend Developer

Summary

The sixth sprint meeting was held briefly to record attendance and maintain consistent sprint documentation. The session served as a check-in rather than a technical discussion, ensuring all present members were accounted for. No new updates on development or paper progress were reported.

- Documentation (Stephen): Initiated the meeting recording and opened the camera for documentation purposes.
- Coordination (Johann): Led the short meeting, confirmed attendance, and concluded the session.
- Engagement (Charles): Participated actively in the conversation, ensuring proper meeting acknowledgment.

Chapters & Topics

No new paper chapters or development topics were discussed. The focus remained on maintaining consistent sprint documentation and verifying team presence.

Assignments (to be completed within Sprint 6)

- Continue development based on previous sprint deliverables (front-end, backend, and UI/UX integration).
- Prepare for Sprint 7 updates and ensure synchronization of ongoing tasks.
- Maintain updated sprint documentation and attendance tracking.

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Notes

- Only three members were present during the meeting (Stephen, Johann, Charles).
- Meeting concluded shortly after confirming attendance and documentation.

Documentation



Noted by,

MARK RISEN CAPARAS

Agile/Scrum Lead

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Lyceum of the Philippines University – Cavite College of Information Technology and Computer Science

ReColor Minutes of the Meeting

October 13, 2025, | 9:00 – 10:00 pm

Agenda

- Sprint Meeting 7

Attendees

1. Mark Risen Caparas – Team Lead
2. Stephen Lacsá – UI/UX Designer
3. Justine Deluana – Honorary Backend Developer
4. Charles Justine Rull – Backend Developer
5. Johann Nikkolai Cepeda – Front-End Developer

Summary

The eighth sprint meeting centered on documentation advancement, front-end development alignment, and backend system testing. The team discussed updates on the manuscript, ongoing integration between modules, and the addition of the Philippine Eye Research Institute's (PERI) admin dashboard as a new system feature.

- **Documentation and Manuscript Progress (Mark):** Reported **80% completion of Chapter 3**, with focus areas on the missing **system flow diagram, algorithm section, model training, and data analysis** components. Overall manuscript completion stands at **60%**, with plans to finalize all pending diagrams and research illustrations.
- **Admin Dashboard Requirement:** Reiterated the need for a **dedicated PERI Admin Dashboard**, intended for research access and result visualization. Developers were advised to incorporate this into the next sprint's tasks.
- **Storyboard and Consultation:** Emphasized the urgency of **completing the storyboard** before proceeding with the **second consultation with advisor Sir Arcel**. The storyboard will serve as a visual and functional guide for upcoming project demonstrations.
- **Front-End and UI/UX Progress (Stephen):** Confirmed that the **foundational planning phase** and all UI revisions have been completed. Shared updates on the **storyboard creation**, aligning it with user flow to maintain design-to-development consistency. Collaboration with Johann on implementation continues.

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- **Back-End Development and System Architecture (Justine):** Delivered another in-depth demonstration of the system's backend, focusing on **CVD simulation accuracy, CNN module enhancements, and integration architecture**. Discussed data logging, unified color processing pipeline, and GPU optimization for real-time testing.

Chapters & Topics

Documentation and Manuscript (Mark)

- Chapter 3: ~80% complete, pending completion of system flow and algorithm-related diagrams.
- Overall manuscript: 60% finalized with remaining sections focused on model training and data analysis.
- Action plan: finalize Chapter 3 and begin testing and evaluation documentation.
- Next step: incorporate feedback from the second consultation after storyboard completion.

UI/UX and Front-End Development (Stephen & Johann)

- Planning and revisions phase completed; UI/UX aligned with final user flow.
- Storyboard designed and synchronized with navigation flow for implementation guidance.
- Transitioning to active front-end development, converting Figma layouts to functional components.
- Collaboration between Stephen and Johann to ensure front-end accuracy and usability.

Backend Development and Integration (Justine & Charles)

- **Integration Status:** 60% completion, with pending incorporation of Charles's CNN code once network access is restored.
- **System Architecture:** Includes TensorFlow-based CNN model for color classification, K-Means clustering, GPU-accelerated processing, and data logging.

Core Modules:

- `main.py` – Entry point for command line and camera initialization.
- `recaller_app.py` – Orchestrator controlling GPU setup, model lifecycle, and unified pipeline.
- `color_model.py` – Handles CNN color analysis and classification.

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- `colorblind_simulator.py` – Implements simulation for Protanopia, Deuteranopia, and Tritanopia.
- `camera_handler.py` – Manages live feed processing and real-time visual adjustments.
- `color_logger.py` – Logs RGB data, AI confidence, and metadata for evaluation.
- `unified_color_pipeline.py` – Merges K-Means, CNN, and Daltonization into a unified enhancement process.

• Demonstration Highlights:

- Real-time visualization of color blindness simulation and correction.
- Adjustable enhancement levels (0.5–3.0x).
- Smooth operation under GPU acceleration with data export capabilities.

Action Items (to be completed within Sprint 8)

Admin Dashboard Implementation:

- Begin development of a **PERI-focused admin dashboard** within the system for data tracking and report generation. (*Developers*)

Storyboard Finalization:

- Finalize the storyboard and prepare for the **advisor consultation** in the upcoming sprint. (*Stephen*)

Code Integration:

- Integrate **Charles's backend code** into the CNN module and perform early-stage validation tests. (*Justine & Charles*)

Documentation Enhancement:

- Complete pending diagrams, finalize the methodology chapter, and begin draft compilation for the testing phase. (*Mark*)

Key Highlights

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- Chapter 3 (Methodology) nearing completion.
- Continued refinement of the CVD simulation algorithm and performance optimization.
- Admin dashboard requirement introduced for PERI research integration.
- Storyboard completion prioritized for the second advisor consultation.

Meeting Adjournment

The meeting adjourned at **10:20 PM**, with the next sprint set to focus on **storyboard completion, backend integration testing, and admin dashboard development**.

Documentation:



Noted by,

MARK RISEN CAPARAS

Agile/Scrum Lead

Appendix C
Initial Survey Questionnaire

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Lyceum of the Philippines University – Cavite College of Information Technology and Computer Science

ReColor Survey Instrument

The purpose of this survey is to gather insights from individuals with Color Vision Deficiency (CVD), ophthalmologists, optometrists, researchers, and related experts to inform the development and evaluation of ReColor: A Mobile Vision Assistive Tool for Color Vision Deficiency Using Adaptive Daltonization and Clustering Algorithms. Specifically, this survey seeks to understand user requirements, current diagnostic and therapeutic practices, technological needs, and opportunities for integrating real-time mobile solutions. The responses will help refine the application to ensure it is practical, accessible, and responsive to the needs of both end-users and professionals in the field of vision care.

General Instructions

1. **Confidentiality:** All responses will be kept confidential and used solely for research purposes.
2. **Completion Time:** The survey will take approximately 10-30 minutes to complete.
3. **Answering Questions:** Please select or provide the most appropriate response based on your experience or perspective. Some questions allow multiple selections, while others may ask for a detailed response.
4. **Optional Questions:** If a question does not apply to you or if you prefer not to answer, feel free to skip it.
5. **Honesty and Accuracy:** Please answer as honestly and accurately as possible to help us gather meaningful data.

Section 1: Respondent Information (For General Participants)

Description: This section helps us understand the background of the respondents.

Layunin: Upang maunawaan ang pinagmulan at kalagayan ng mga kalahok.

1. What is your role? (Select the most appropriate response)
 - a. Student
 - b. Working Professional
 - c. Other (please specify)

Ano ang inyong katayuan? (Piliin ang naaangkop na sagot)

- a. Estudyante
- b. Nagtatrabaho
- c. Iba pa (pakispecify)

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may Color Vision Deficiency (CVD) at higit pang magpapalawak sa larangan ng pananaliksik sa kalusugan ng paningin.

Muli, taos-pusong pasasalamat sa inyong kooperasyon at suporta para sa pananaliksik na ito.

Kung ikaw ay may katanungan maaring ikontak ang mga sumusunod:

MARK RISEN B. CAPARAS

Email: mark.caparas@lpnetwork.edu.ph

Phone: +63 967 514 5809

CHARLES JUSTINE B. RULL

Email: charles.rull@lpnetwork.edu.ph

Phone: +639 9889 8407

MR. JOHANN NIKKOLAI L. CEPEDA

Email: johann.cepeda@lpnetwork.edu.ph

Phone: +639 205 6752

MR. STEPHEN L. LACSA

Email: stephen.lacsa@lpnetwork.edu.ph

Phone: +639 563 17375

We look forward to hearing your insights!

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2. Do you have a diagnosed Color Vision Deficiency (CVD)?
- Yes
 - No
 - Not diagnosed, but I believe I have difficulty with colors

Mayroon ka bang opisyal na diagnosis ng CVD?

- Oo
- Wala
- Wala, ngunit naniniwala akong may problema ako sa kulay

3. What device do you primarily use?
- Android phone
 - iPhone
 - Tablet
 - Other

Anong device ang pangunahing ginagamit mo?

- Android phone
- iPhone
- Tablet
- Iba pa

Section 2: Experiences with Color Vision Deficiency (For General Participants)

Description: This section identifies challenges and coping strategies related to CVD.

Layunin: Upang matukoy ang mga hamon at paraan ng pagharap ng mga taong may CVD.

1. In what situations do you find color vision most challenging? (Select all that apply)
- Reading maps or charts
 - Identifying traffic lights or road signs
 - Academic activities (labs, diagrams, graphs)
 - Workplace tasks (design, coding, quality inspection)
 - Daily tasks (clothes matching, cooking, shopping)
 - Other

Sa anong sitwasyon ka nahihirapan dahil sa CVD? (Pumili ng lahat ng naaangkop)

2. Have you used any tools to cope with CVD?
- None
 - Glasses with tinted lenses
 - Mobile applications
 - Other



May ginagamit ka bang tulong para sa CVD?

3. Would you find a mobile app that provides both screening (Ishihara Test) and real-time color correction useful?
- Yes
 - No
 - Maybe

Makakatulong ba sa iyo ang isang mobile app na may screening (Ishihara Test) at real-time color correction?

Section 3: Technical and Expert Perspectives (For Ophthalmologists, Optometrists, and Researchers)

Description: This section explores diagnostic tools, therapies, and technology for CVD.

Layunin: Upang suriin ang kasalukuyang kaalaman at teknolohiya para sa CVD.

1. Can all types of CVD be corrected, or are certain types (e.g., protanopia, deuteranopia, tritanopia) more responsive to interventions?

Lahat ba ng uri ng CVD ay maaaring maitama, o may mga partikular na uri na mas tumutugon sa interbensyon?

2. What methods or interventions are currently used for managing or treating CVD?

Anu-ano ang kasalukuyang pamamaraan o interbensyon na ginagamit sa pamamahala o paggamot ng CVD?

3. How effective are these interventions in improving daily visual tasks?

Gaano kaepektibo ang mga interbensyong ito sa pagpapabuti ng araw-araw na gawaing biswal?

4. Are there limitations in existing diagnostic tools (e.g., Ishihara, Farnsworth D-15) in detecting severity or subtype of CVD?

Mayroon bang limitasyon ang kasalukuyang diagnostic tools sa pagtukoy ng antas o subtype ng CVD?

5. What role does digital technology (e.g., mobile apps, AR/VR systems, daltonization) currently play in assisting CVD patients?

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Ano ang papel ng digital technology sa pagtulong sa mga pasyenteng may CVD?

6. How do assistive tools like tinted lenses or mobile apps compare in effectiveness with clinical therapies?

Paano ikinukumpara ang bisa ng assistive tools sa klinikal na therapies?

7. Could machine learning or adaptive algorithms be used to track and personalize therapy progress for CVD patients?

Maaari bang gamitin ang machine learning upang masubaybayan at i-personalize ang therapy ng mga pasyenteng may CVD?

8. Can personalized therapy approaches lead to better outcomes compared to standardized methods?

Maaari bang mas maging mabisa ang personalized therapy kaysa sa standardized methods?

9. Are there observable age-related differences in responsiveness to therapies?

Mayroon bang pagkakaiba sa pagtugon sa therapy batay sa edad?

10. What environmental factors (e.g., lighting, screen type) affect the success of therapies or assistive tools?

Anong mga salik sa kapaligiran ang nakakaapekto sa tagumpay ng therapies o assistive tools?

11. Are there ethical concerns with artificially modifying or enhancing human color perception?

May mga etikal bang isyu sa artipisyal na pagbabago o pagpapahusay ng kakayahan ng tao sa pagkilala ng kulay?



Section 4: General Feedback

Description: Open-ended items to gather final input.

Layunin: Upang makakuha ng pangkalahatang mungkahi at pananaw.

1. What advice can you give to improve ReColor's usefulness for CVD individuals in the Philippines?

Anong payo ang maaari mong ibigay upang mas maging kapaki-pakinabang ang ReColor para sa mga taong may CVD sa Pilipinas?

2. What features would you like to see in future developments of ReColor?

Anong mga karagdagang tampok ang nais mong makita sa mga susunod na bersyon ng ReColor?

Thank You for Your Participation

Thank you very much for your time and participation in completing this survey. Your responses are highly valuable in deepening our study and contributing meaningful results to the project *ReColor: A Mobile Vision Assistive Tool for Color Vision Deficiency Using Adaptive Daltonization and Clustering Algorithms*.

Through your honest and thoughtful answers, you are helping us develop a more efficient and effective solution that can benefit individuals with Color Vision Deficiency (CVD) and further advance research in the field of vision health.

Once again, we sincerely appreciate your cooperation and support for this study.

If you have any questions about this study, please contact:

Maraming Salamat sa Iyong Partisipasyon

Maraming salamat po sa inyong oras at pakikiisa sa pagsagot ng survey na ito. Ang inyong mga tugon ay napakahalaga upang mas mapalalim ang aming pag-aaral at makapagbigay ng makabuluhang resulta para sa proyektong ReColor: A Mobile Vision Assistive Tool for Color Vision Deficiency Using Adaptive Daltonization and Clustering Algorithms.

Sa pamamagitan ng inyong tapat at makabuluhang mga sagot, makatutulong po kayo sa pagbuo ng isang mas episyente at epektibong solusyon na makikinabang ang mga taong

Appendix D

Narrative



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ReColor
ReColor Logo Narrative



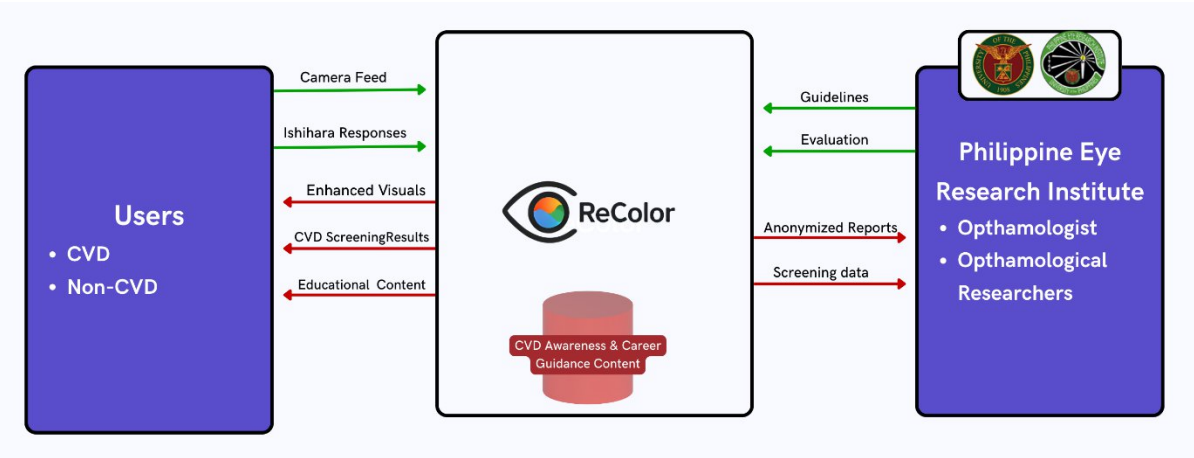
Recolor Official Logo (2025)

The ReColor logo was created in May 2025 by Mark Risen B. Caparas during the early conceptualization stage of the project in the Research Methods in Computing course. It symbolizes the mission of enhancing vision accessibility for individuals with Color Vision Deficiency (CVD). The design features a spectrum of colors arranged in a circular form, representing inclusivity and the full range of human color perception. The circular motif also draws inspiration from Ishihara test plates, a universal screening tool for CVD, and reflects continuity in vision care from diagnosis to assistance.

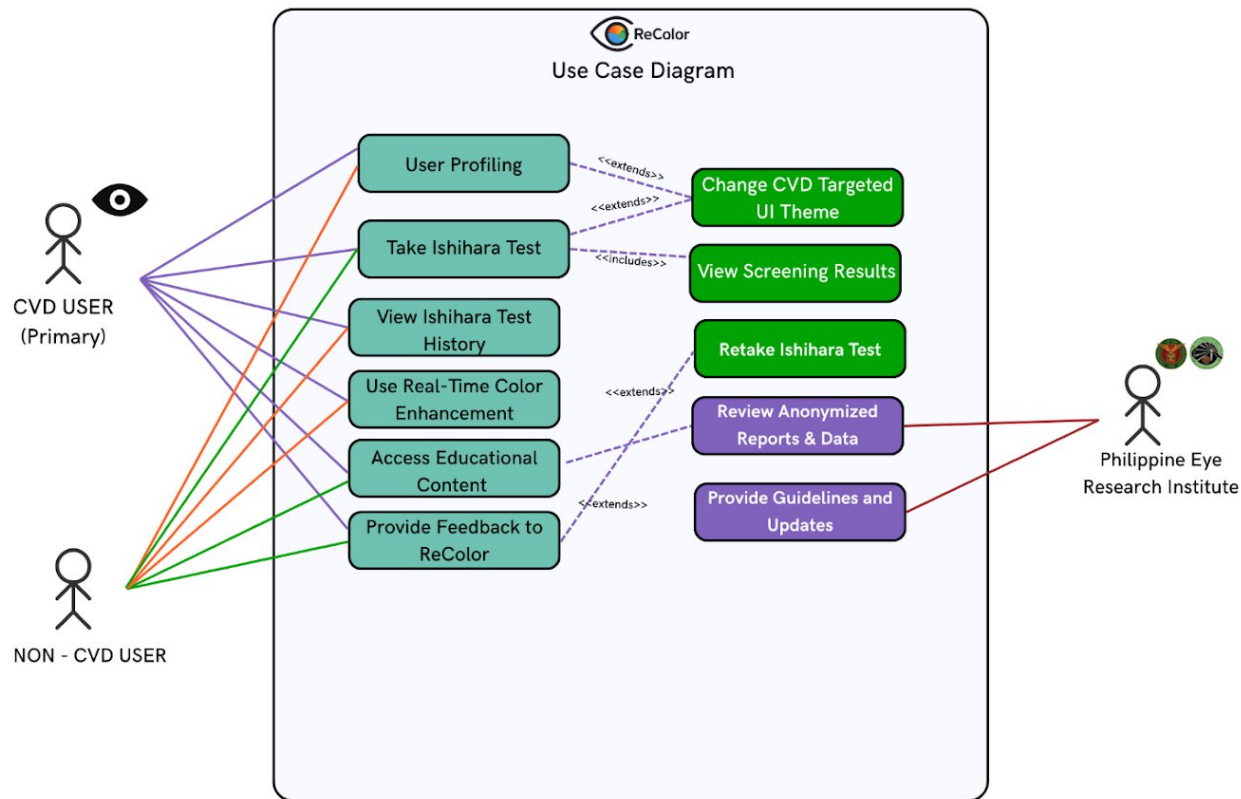
The gradient transition within the logo signifies the adaptive daltonization and clustering algorithms that form the core of ReColor, dynamically remapping colors to improve real-time perception. Its clean, modern typography ensures readability and aligns with accessibility standards, emphasizing the project's user-centered design. Additionally, contrasting hues were deliberately chosen to demonstrate colorblind-friendly palette practices, ensuring visibility even for users with red-green anomalies.

ReColor logo functions not only as a visual identifier but also as a symbolic representation of innovation, accessibility, and alignment with Republic Act No. 11358 or the National Vision Screening Act. By linking its creation to the early conceptual stages of the project, the logo underscores ReColor's enduring commitment to bridging the gap between clinical vision screening and practical mobile assistance, contributing to national efforts for inclusive eye health in the Philippines.

Appendix E
Context Diagram



Appendix F
Use Case Diagram



Appendix G
Evaluation Matrix for ReColor

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Module	Metric	Formula / Basis	Purpose	Interpretation
Ishihara Module	Accuracy (%)	$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$	To measure the proportion of correctly classified cases compared with the standard Ishihara test.	Higher values indicate closer alignment with clinical Ishihara test outcomes.
	Sensitivity	$Sensitivity = \frac{TP}{TP + FN}$	To evaluate the system's ability to correctly detect users with CVD.	High sensitivity reduces risk of missed diagnoses (false negatives).
	Specificity	$Specificity = \frac{TN}{TN + FP}$	To evaluate the system's ability to correctly identify non-CVD users.	High specificity reduces risk of misdiagnosing normal users as CVD.
	Cohen's Kappa (κ)	$\kappa = \frac{po - pe}{1 - pe}$	To assess agreement with expert classification, adjusting for chance.	$\kappa > 0.80$ indicates strong reliability of the Ishihara module.

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Color
Correction
Module
(Daltonization + K-Means)

ΔE
(CIEDE2000)

$$\Delta E = \sqrt{\left(\frac{\Delta L^*}{K_L S_L}\right)^2 + \left(\frac{\Delta C^*}{K_C S_C}\right)^2 + \left(\frac{\Delta B^*}{K_B S_B}\right)^2 + R_T \left(\frac{\Delta C^*}{K_C S_C}\right) \left(\frac{\Delta B^*}{K_B S_B}\right)}$$

To quantify perceptual differences between original and corrected colors.

Lower ΔE values indicate higher perceptual accuracy, closer to normal vision perception.

System
Responsiveness

Frame rate (fps) and latency (ms)

To measure real-time feasibility of correction on mobile devices.

Higher fps and lower latency indicate better user experience and responsiveness.

CNN-Based
Color
Identifier
Module

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

To measure the proportion of correctly predicted colors among all predictions.

High precision ensures correct classification of identified colors.

Recall
(Sensitivity)

$$\text{Recall} = \frac{TP}{TP + FN}$$

To measure the ability to detect all true instances of a given color.

High recall minimizes missed classifications.

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F1 Score

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

To provide a balanced measure of precision and recall.

Higher F1 indicates strong classification even with class imbalance.

Confusion Matrix

Matrix of TP, TN, FP, FN per class

To visualize class-level performance and identify misclassifications.

Clearer picture of strengths/weaknesses in CNN predictions.

Appendix H
Philippine Eye Research Institute Consultation Narrative Reports

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Lyceum of the Philippines University – Cavite
College of Information Technology and Computer Science

ReColor
1st Consultation with the Philippine Eye Research Institute (PERI)

Date: August 28, 2025

Location: Philippine Eye Research Institute,

Mode: Face-to-Face Interview

Participants:

1. Mark Risen Caparas – Team lead
2. Johann Nikkolai Cepeda - Front-End Developer
3. Justine Deluana – Honorary Backend Developer
4. RES-001 (Pseudonym), Researcher from the Philippine Eye Research Institute

Purpose:

To gather expert insights on the scientific, diagnostic, and ethical foundations of the ReColor system, focusing on the clinical context of Color Vision Deficiency (CVD) and the practical considerations for integrating Ishihara-based screening into a mobile application.

Summary of Discussion:

The consultation emphasized that over 90% of CVD cases are genetic and lifelong, though manageable through adaptation and environmental cues. The expert clarified that while optical aids (e.g., tinted spectacles) enhance color contrast, they do not restore color perception. Gene therapy remains experimental and inaccessible for most patients. Locally, adaptive coping mechanisms—such as recognizing color positions in traffic lights—are common due to the lack of formal therapy or interventions.

Regarding digital diagnostic tools, the expert affirmed that the Ishihara test remains the gold standard but has limitations when displayed digitally due to device calibration differences, brightness, and screen protectors. Standardization across devices was recommended to improve reliability. The expert advised including a disclaimer stating that digital Ishihara tests are screening tools only and not substitutes for professional diagnosis.

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Additionally, the discussion highlighted that **protanopia and deuteranopia** are the most prevalent forms of CVD in the Philippines. Clinical validation of ReColor was considered feasible but constrained by sample availability, given the low number of diagnosed cases. Suggested recruitment sources include seafarer medical examinations and aviation screenings, where color perception testing is routine.

The expert commended ReColor's adaptive correction approach and recommended exploring device calibration protocols and user-specific configuration to enhance accuracy. The meeting concluded with agreement to maintain ethical compliance, particularly in data handling, anonymization, and adherence to PERI's guidance on research involving vision testing.

Note:

The interview was conducted under informed consent and with strict adherence to data privacy regulations. Participant anonymity was maintained using a pseudonym (RES-001). A signed consent form is filed separately and referenced via participant code.

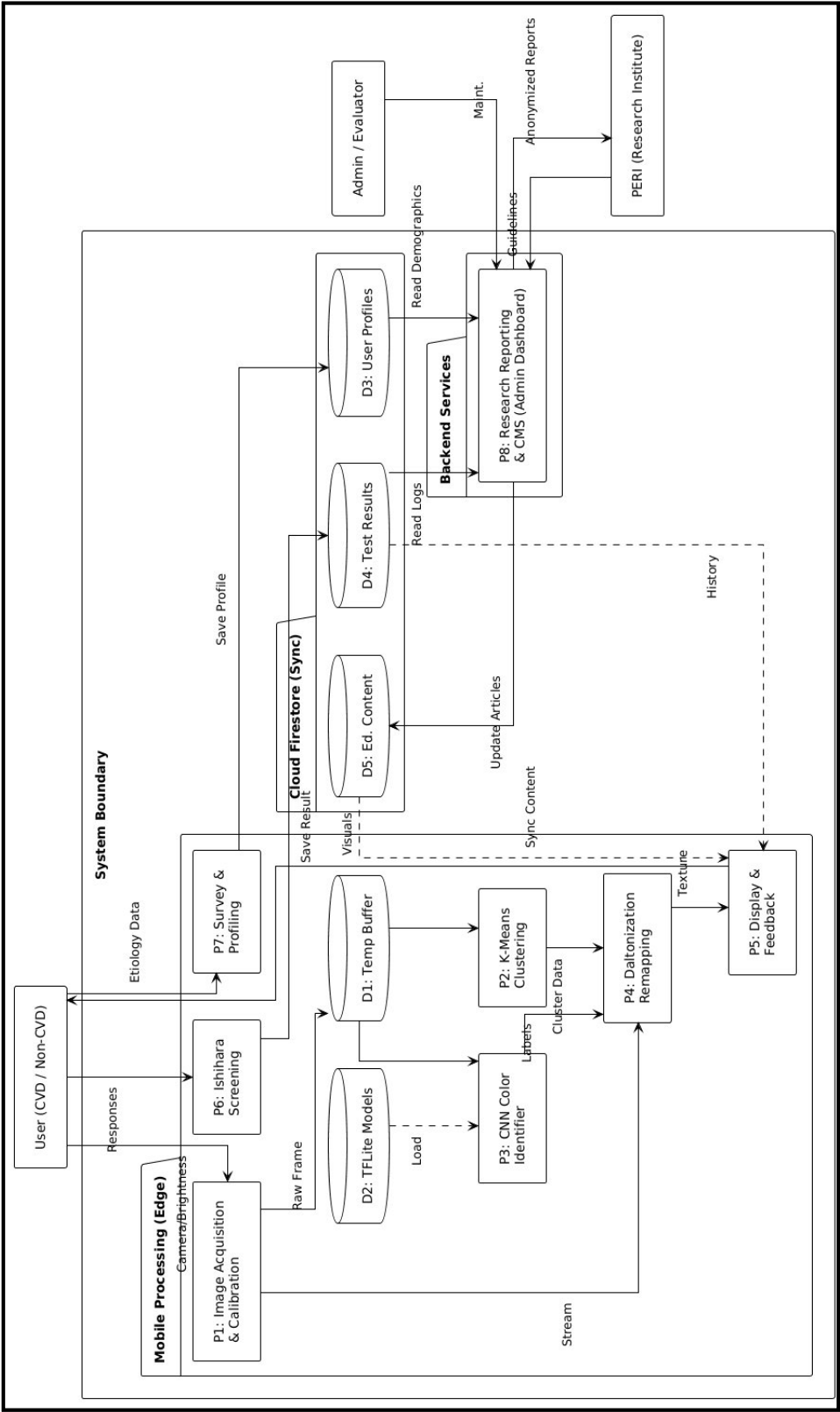
Documentation:



Documented by:

Mark Risen Caparas
Team Lead, ReColor

Appendix I
Data Flow Diagram (Level 1)



Appendix J
Relevant Source Code

K-Means Clustering

```
import os
import numpy as np
from PIL import Image
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Define dataset path and color categories
dataset_path =
"C:\\xampp\\htdocs\\Evala\\ReColor\\ReColor-Backend\\color_dataset_rgb_hsv"
color_families = ['red', 'green', 'blue']

dataset_pixels = []
labels = []

# Load and preprocess dataset
for color_name in color_families:
    folder_path = os.path.join(dataset_path,
    color_name)
    for file in os.listdir(folder_path):
        img_path = os.path.join(folder_path, file)
        img =
Image.open(img_path).convert("RGB").resize((32,
32))
        pixels = np.array(img).reshape(-1, 3)
        dataset_pixels.append(pixels)
        labels.extend([color_name] * len(pixels))

dataset_pixels = np.vstack(dataset_pixels)
print("Dataset pixel array shape:",
dataset_pixels.shape)

# Apply K-Means clustering
k = 3 # Initial number of clusters (expand to 9 in
later experiments)
kmeans = KMeans(n_clusters=k,
random_state=42)
kmeans.fit(dataset_pixels)

print("K-Means centroids:\n",
kmeans.cluster_centers_.astype(int))

plt.title("Original")
plt.imshow(image)
plt.axis("off")

plt.subplot(1,2,2)
plt.title("Clustered (R/G/B Families)")
plt.imshow(clustered_img)
plt.axis("off")
```

```
# Visualize clustering result
image_path =
"C:\\xampp\\htdocs\\Evala\\ReColor\\test.jpg"
image = Image.open(image_path).convert("RGB")
pixels = np.array(image)
h, w, _ = pixels.shape
pixels_2d = pixels.reshape(-1, 3)

cluster_labels = kmeans.predict(pixels_2d)
clustered_img =
kmeans.cluster_centers_[cluster_labels].reshape(h, w,
3).astype(np.uint8)

plt.figure(figsize=(10,5))
plt.subplot(1,2,1)
```

CNN

Listing 1. Model Architecture (EfficientNetB0 backbone + custom head)

```
import tensorflow as tf
from tensorflow.keras import layers, models

NUM_CLASSES = 19

def build_color_model(image_size=224,
num_classes=NUM_CLASSES, dropout=0.3):
    inputs = layers.Input(shape=(image_size,
image_size, 3))
    backbone = tf.keras.applications.EfficientNetB0(
include_top=False, weights="imagenet",
input_tensor=inputs, pooling="avg"
)
    x = layers.Dropout(dropout)(backbone.output)
    outputs = layers.Dense(num_classes,
activation="softmax")(x)
    model = models.Model(inputs, outputs)
    return model
```

Listing 2. MacroF1 Keras Callback (early checkpointing)

```
import numpy as np
from sklearn.metrics import f1_score
import tensorflow as tf

class MacroF1Callback(tf.keras.callbacks.Callback):
    def __init__(self, val_data, patience=3,
save_path="best_macro_f1.keras"):
        super().__init__()
        self.val_data = val_data
        self.patience = patience
```


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```

self.val_data = val_data
self.patience = patience
self.save_path = save_path
self.best_f1 = -1.0
self.wait = 0

def on_epoch_end(self, epoch, logs=None):
    y_true, y_pred = [], []
    for x_batch, y_batch in self.val_data:
        probs = self.model.predict(x_batch,
        verbose=0)
        y_true.extend(np.argmax(y_batch, axis=1))
        y_pred.extend(np.argmax(probs, axis=1))
        macro_f1 = f1_score(y_true, y_pred,
        average="macro")
        logs["val_macro_f1"] = macro_f1
        if macro_f1 > self.best_f1 + 1e-6:
            self.best_f1 = macro_f1
            self.wait = 0
            self.model.save(self.save_path)
        else:
            self.wait += 1
            if self.wait >= self.patience:
                self.model.stop_training = True

```

Listing 3. Class-Conditional Augmentation + Sample Weighting

```

import tensorflow as tf

# Example per-class weights emphasizing navy
CLASS_NAMES = [

    "beige_tan", "black", "blue", "brown", "cyan", "gold", "gray", "green", "lime",

    "magenta", "navy", "orange", "pink", "purple", "red", "silver", "teal", "white", "yellow"
]
CLASS_WEIGHTS = {name: 1.0 for name in CLASS_NAMES}
CLASS_WEIGHTS["navy"] = 15.0 # boosted to remediate navy→blue confusion

def augment(image, label_index):
    image = tf.image.random_flip_left_right(image)
    image = tf.image.random_brightness(image, 0.08)
    image = tf.image.random_contrast(image, 0.9, 1.1)
    # Class-conditional tweaks (navy darken; blue slight brighten)
    navy_idx = CLASS_NAMES.index("navy")
    blue_idx = CLASS_NAMES.index("blue")

```

```

    navy_idx = CLASS_NAMES.index("navy")
    blue_idx = CLASS_NAMES.index("blue")
    cond_navy = tf.equal(label_index, navy_idx)
    cond_blue = tf.equal(label_index, blue_idx)
    image = tf.where(cond_navy, image * 0.90, image)
    image = tf.where(cond_blue,
    tf.clip_by_value(image * 1.05, 0, 255), image)
    return image

```

```

def preprocess(image, label):
    image = tf.image.resize(image, (224, 224))
    image = augment(image, tf.argmax(label))
    image =
    tf.keras.applications.efficientnet.preprocess_input(image)
    # Attach sample weight
    class_name = CLASS_NAMES[tf.argmax(label)]
    weight =
    CLASS_WEIGHTS[class_name.numpy().decode()]
    return image, label, tf.cast(weight, tf.float32)

```

```

def make_dataset(root, batch=32):
    ds =
    tf.keras.preprocessing.image_dataset_from_directory(
        root, image_size=(224, 224),
        label_mode="categorical", shuffle=True
    )
    ds = ds.map(preprocess,
    num_parallel_calls=tf.data.AUTOTUNE)
    return ds.prefetch(tf.data.AUTOTUNE)

```

Listing 4. Model Compilation and Training Loop

```

train_ds = make_dataset("data/v5_split/train")
val_raw =
tf.keras.preprocessing.image_dataset_from_directory(
    "data/v5_split/val", image_size=(224, 224),
    label_mode="categorical", shuffle=False
)
val_ds = val_raw.map(
    lambda x, y:
    (tf.keras.applications.efficientnet.preprocess_input(x),
    y),
    num_parallel_calls=tf.data.AUTOTUNE
)

model = build_color_model()
model.compile(

    optimizer=tf.keras.optimizers.AdamW(learning_rate=
    3e-4, weight_decay=1e-4),
    loss="categorical_crossentropy",

```

```
optimizer=tf.keras.optimizers.AdamW(learning_rate=3e-4, weight_decay=1e-4),
    loss="categorical_crossentropy",
    metrics=["accuracy"]
)

macro_cb = MacroF1Callback(val_ds)
history = model.fit(
    train_ds,
    epochs=12,
    validation_data=val_ds,
    callbacks=[macro_cb],
    class_weight=None, # using per-sample weights
    from dataset
)
```

Listing 5. TFLite Export (FP32 and FP16)

```
import tensorflow as tf

def export_tflite(keras_path, out_fp32, out_fp16):
    model = tf.keras.models.load_model(keras_path)
    converter =
    tf.lite.TFLiteConverter.from_keras_model(model)
    # FP32
    tflite_fp32 = converter.convert()
    open(out_fp32, "wb").write(tflite_fp32)
    # FP16
    converter.optimizations =
    [tf.lite.Optimize.DEFAULT]
    converter.target_spec.supported_types =
    [tf.float16]
    tflite_fp16 = converter.convert()
    open(out_fp16, "wb").write(tflite_fp16)

export_tflite("best_macro_f1.keras",
    "color_model_19cls_fp32.tflite",
    "color_model_19cls_fp16.tflite")
```

Listing 6. Inference Helper with Achromatic Gate

```
import numpy as np
import tensorflow as tf
from PIL import Image

ACHROMATIC_S_THRESH = 0.08
ACHROMATIC_V_BLACK = 0.20
ACHROMATIC_V_WHITE = 0.85

def load_labels(path="labels.json"):
    import json
    with open(path, "r") as f:
```

```
LABELS = load_labels()
```

```
def achromatic_gate(rgb_uint8):
    hsv = Image.fromarray(rgb_uint8).convert("HSV")
    arr = np.array(hsv, dtype=np.float32)
    s_mean = arr[...,1].mean() / 255.0
    v_mean = arr[...,2].mean() / 255.0
    if s_mean < ACHROMATIC_S_THRESH and
    v_mean < ACHROMATIC_V_BLACK:
        return "black"
    if s_mean < ACHROMATIC_S_THRESH and
    v_mean > ACHROMATIC_V_WHITE:
        return "white"
    return None
```

```
def preprocess_image(path):
    img =
    Image.open(path).convert("RGB").resize((224, 224))
    arr = np.array(img).astype(np.float32)
    arr =
    tf.keras.applications.efficientnet.preprocess_input(arr)
    return np.expand_dims(arr, 0)
```

```
def predict_tflite(image_path, tflite_path):
    interpreter =
    tf.lite.Interpreter(model_path=tflite_path)
    interpreter.allocate_tensors()
    input_details = interpreter.get_input_details()
    output_details = interpreter.get_output_details()
```

```
    gate_label =
    achromatic_gate(np.array(Image.open(image_path).c
    onvert("RGB").resize((224,224))))
    if gate_label:
        return {"label": gate_label, "index": -1, "gate":
    True}
```

```
    x = preprocess_image(image_path)
    interpreter.set_tensor(input_details[0]["index"], x)
    interpreter.invoke()
    probs =
    interpreter.get_tensor(output_details[0]["index"])[0]
    idx = int(np.argmax(probs))
    return {"label": LABELS[idx], "index": idx,
    "probs": probs.tolist()}
```

Daltonization

```
// src/shaders/daltonize.ts
export const DALTONIZE_SKSL = `
uniform shader u_Image;
uniform float u_Strength; // correction gain k
in [0,1]
uniform float3x3 u_RGB_TO_LMS; //
M_RGB→LMS
uniform float3x3 u_LMS_TO_RGB; //
M_LMS→RGB
uniform float3x3 u_SIMULATION; // M_Sim
uniform float3x3 u_CORRECTION; // M_Corr
uniform int u_OutputSimulated; // 1 = LMS_Sim,
0 = LMS_Corr
```

```
float3 srgbToLinear(float3 c) {
    float3 cutoff = step(float3(0.04045), c);
    float3 low = c / 12.92;
    float3 high = pow((c + 0.055) / 1.055, 2.4);
    return mix(low, high, cutoff);
}
```

```
float3 linearToSrgb(float3 c) {
    float3 cutoff = step(float3(0.0031308), c);
    float3 low = 12.92 * c;
    float3 high = 1.055 * pow(c, 1.0/2.4) - 0.055;
    return mix(low, high, cutoff);
}
```

```
half4 main(float2 xy) {
    half4 inC = u_Image.eval(xy);
    float3 rgb = srgbToLinear(float3(inC.rgb)); //
    Linearization
```

```
    float3 LMS_Orig = u_RGB_TO_LMS * rgb;
    // LMS_Orig
    float3 LMS_Sim = u_SIMULATION *
    LMS_Orig; // LMS_Sim = M_Sim ·
    LMS_Orig
    float3 E = LMS_Orig - LMS_Sim; //
    E = LMS_Orig - LMS_Sim
    float3 comp = u_CORRECTION * E;
    // M_Corr · E
    float3 LMS_Corr = LMS_Orig + u_Strength *
    comp; // LMS_Corr = LMS_Orig + k · (M_Corr ·
    E)
```

```
    float3 rgbSimLin = u_LMS_TO_RGB *
    LMS_Sim;
    float3 rgbCorrLin = u_LMS_TO_RGB *
    LMS_Corr;
```

```
    float3 outLin = (u_OutputSimulated == 1) ?
```

```
    float3 outLin = (u_OutputSimulated == 1) ?
    rgbSimLin : rgbCorrLin;
    outLin = clamp(outLin, 0.0, 1.0);
```

```
    float3 srgbOut = linearToSrgb(outLin);
    return half4(half3(srgbOut), inC.a);
}
```

LMS Transform and CVD Matrices (HPE + Viénot/Brettel-Style)

```
// src/constants/matrices.ts
```

```
// Hunt-Pointer-Estevéz RGB→LMS and
LMS→RGB (column-major for SKSL)
```

```
export const RGB_TO_LMS: number[] = [
    17.8824, 3.45565, 0.02996,
    43.5161, 27.1554, 0.18431,
    4.11935, 3.86714, 1.46709,
];
```

```
export const LMS_TO_RGB: number[] = [
    0.0809, -0.1305, 0.1167,
    -0.0102, 0.0540, -0.1136,
    -0.0004, -0.0041, 0.6935,
];
```

```
// Protanopia (red-blind)
```

```
export const SIM_PROTAN: number[] = [
    0.0, 0.0, 0.0,
    2.02344, 1.0, 0.0,
    -2.52581, 0.0, 1.0,
];
```

```
export const CORR_PROTAN: number[] = [
    0.0, 0.7, 0.7,
    0.0, 1.0, 0.0,
    0.0, 0.0, 1.0,
];
```

```
// Deuteranopia (green-blind)
```

```
export const SIM_DEUTAN: number[] = [
    1.0, 0.494207, 0.0,
    0.0, 0.0, 0.0,
    0.0, 1.24827, 1.0,
];
```

```
export const CORR_DEUTAN: number[] = [
    1.0, 0.0, 0.0,
    0.7, 0.0, 0.7,
    0.0, 0.0, 1.0,
];
```

```
// Tritanopia (blue-blind)
```

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```
// Tritanopia (blue-blind)
export const SIM_TRITAN: number[] = [
  1.0, 0.0, -0.395913,
  0.0, 1.0, 0.801109,
  0.0, 0.0, 0.0,
];

export const CORR_TRITAN: number[] = [
  1.0, 0.0, 0.0,
  0.0, 1.0, 0.0,
  0.7, 0.7, 0.0,
];

export type CVDType = 'protan' | 'deutan' | 'tritan';

export const MATRICES_BY_TYPE:
Record<CVDType, {
  RGB_TO_LMS: number[];
  LMS_TO_RGB: number[];
  SIMULATION: number[];
  CORRECTION: number[];
}> = {
  protan: { RGB_TO_LMS, LMS_TO_RGB,
SIMULATION: SIM_PROTAN, CORRECTION:
CORR_PROTAN },
  deutan: { RGB_TO_LMS, LMS_TO_RGB,
SIMULATION: SIM_DEUTAN, CORRECTION:
CORR_DEUTAN },
  tritan: { RGB_TO_LMS, LMS_TO_RGB,
SIMULATION: SIM_TRITAN, CORRECTION:
CORR_TRITAN },
};

3. CPU Reference Implementation (Used for Static Figures)

// Core CPU daltonization pipeline (simplified from
scripts/export_matrix.js)

const HPE_RGB_TO_LMS = [
  0.31399022, 0.63951294, 0.04649755,
  0.15537241, 0.75789446, 0.08670142,
  0.01775239, 0.10944209, 0.87256922,
];

const HPE_LMS_TO_RGB = [
  5.47221206, -4.6419601, 0.16963708,
  -1.1252419, 2.29317094, -0.1678952,
  0.02980165, -0.19318073, 1.16364789,
];

function srgbToLinear(u) {
  const c = u / 255;
  if (c <= 0.04045) return c / 12.92;
  return Math.pow((c + 0.055) / 1.055, 2.4);
}
```

```
function srgbToLinear(u) {
  const c = u / 255;
  if (c <= 0.04045) return c / 12.92;
  return Math.pow((c + 0.055) / 1.055, 2.4);
}

function linearToSrgb(u) {
  const c = Math.max(0, Math.min(1, u));
  if (c <= 0.0031308) return Math.round(255 *
12.92 * c);
  return Math.round(255 * (1.055 * Math.pow(c,
1/2.4) - 0.055));
}

function mul3x3(M, v) {
  return [
    M[0]*v[0] + M[1]*v[1] + M[2]*v[2],
    M[3]*v[0] + M[4]*v[1] + M[5]*v[2],
    M[6]*v[0] + M[7]*v[1] + M[8]*v[2],
  ];
}

// SIM and CORR as in matrices.ts, but row-major:
const SIM = {
  protan: [ 0.0, 1.05118294, -0.05116099,
    0.0, 1.0, 0.0,
    0.0, 0.0, 1.0 ],
  deutan: [ 1.0, 0.0, 0.0,
    0.9513092, 0.0, 0.04866992,
    0.0, 0.0, 1.0 ],
  tritan: [ 1.0, 0.0, 0.0,
    0.0, 1.0, 0.0,
    -0.86744736, 1.86727089, 0.0 ],
};

const CORR = {
  protan: [ 0.0, 0.0, 0.0,
    0.7, 1.0, 0.0,
    0.7, 0.0, 1.0 ],
  deutan: [ 1.0, 0.7, 0.0,
    0.0, 0.0, 0.0,
    0.0, 0.7, 1.0 ],
  tritan: [ 1.0, 0.0, 0.7,
    0.0, 1.0, 0.7,
    0.0, 0.0, 0.0 ],
};

function daltonizePixel_srgb(R, G, B, type, k /*
strength */) {
  // Linearization
  const r = srgbToLinear(R);
  const g = srgbToLinear(G);
  const b = srgbToLinear(B);
```

```

const r = srgbToLinear(R);
const g = srgbToLinear(G);
const b = srgbToLinear(B);

// LMS_Orig = M_RGB→LMS · RGB_Linear
const LMS_Orig = mul3x3(HPE_RGB_TO_LMS,
[r, g, b]);

// LMS_Sim = M_Sim · LMS_Orig
const Msim = SIM[type];
const LMS_Sim = mul3x3(Msim, LMS_Orig);

// E = LMS_Orig - LMS_Sim
const E = [
  LMS_Orig[0] - LMS_Sim[0],
  LMS_Orig[1] - LMS_Sim[1],
  LMS_Orig[2] - LMS_Sim[2],
];

// LMS_Corr = LMS_Orig + k · (M_Corr · E)
const Mcorr = CORR[type];
const comp = mul3x3(Mcorr, E);
const LMS_Corr = [
  LMS_Orig[0] + k * comp[0],
  LMS_Orig[1] + k * comp[1],
  LMS_Orig[2] + k * comp[2],
];

// Back to linear RGB, then to sRGB
const rgbCorrLin = mul3x3(HPE_LMS_TO_RGB,
LMS_Corr);
const outR = linearToSrgb(rgbCorrLin[0]);
const outG = linearToSrgb(rgbCorrLin[1]);
const outB = linearToSrgb(rgbCorrLin[2]);

return [outR, outG, outB];
}

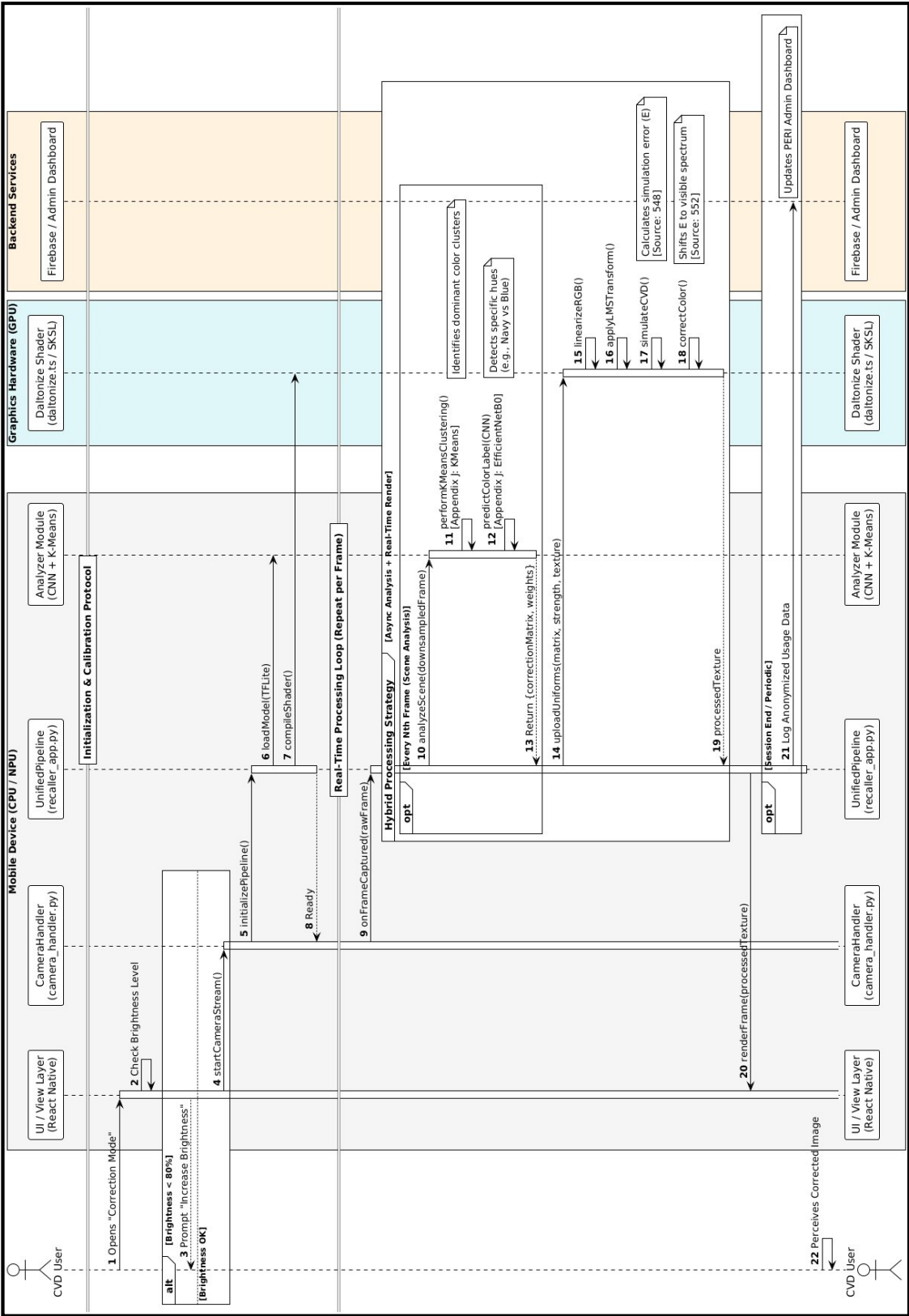
```

Appendix K
Daltonization Transformation Matrices

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Appendix L
Sequence Diagram



Appendix M
Entity Relationship Diagram

